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RESEARCH ARTICLE

Mitigating Bias in AI Model Using eXplainable AI in Terms of Hiring Process in the Industry

NOIR ALSUBAIE AND NOURA ALEISA
College of Computing and Informatics, Saudi Electronic University, Riyadh 13316, Saudi Arabia
Corresponding author: Noura Aleisa (n.aleisa@seu.edu.sa)

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ABSTRACT The increasing adoption of Artificial Intelligence (AI) in recruitment has introduced substantial ethical concerns, particularly regarding the propagation of unintended biases that may reinforce or exacerbate existing employment disparities. In response to these challenges, this study proposes an Explainable Artificial Intelligence (XAI) approach that transitions from opaque "black-box" algorithms to transparent "glass-box" frameworks, enabling greater interpretability and accountability. Specifically, we employ SHapley Additive exPlanations (SHAP) to examine and interpret decisions made by widely used machine learning models—namely Random Forest, XGBoost, and LightGBM—within the context of AI-driven hiring systems. Through extensive empirical analysis, SHAP values were utilized to uncover key features, such as gender, nationality, and recreational preferences, that contributed disproportionately to biased outcomes. These features were iteratively reviewed, modified, or excluded to construct fairer models. Following refinement, the updated models demonstrated a notable increase in classification accuracy, rising from approximately 78% to 85%. In addition, precision, recall, and F1-score metrics improved across all model variants. Most significantly, fairness metrics, measured through demographic parity, showed an approximate 20% improvement, increasing from 0.70 to 0.90, indicating a substantial reduction in discriminatory bias. These results provide strong empirical support for integrating XAI techniques in recruitment pipelines, demonstrating that fairness and accuracy need not be mutually exclusive. This study underscores the practical viability of explainable methods in bias-sensitive domains and offers a replicable framework for ethically aligned model development. Ultimately, our findings advocate for adopting XAI methodologies as an essential safeguard in ensuring equitable, transparent, and socially responsible deployment of AI technologies in employment decision-making.

INDEX TERMS eXplainable artificial intelligence, XGBoost, recruitment, bias, XAI, random forest, LightGBM.

I. INTRODUCTION

As Artificial Intelligence (AI) technology rapidly evolves, its deployment in recruitment strategies has become increasingly prevalent, promising enhanced efficiency, decision-making capabilities, and talent acquisition optimization. However, this technological advancement brings to light significant concerns regarding inherent biases within AI models, which could potentially exacerbate or introduce new forms of employment disparities. The paper aims to demonstrate how the integration of eXplainable AI (XAI) can transform opaque AI systems into transparent 'glass boxes' models, fostering a more equitable hiring landscape. Through a methodical investigation, including a comparative analysis of various AI algorithms and the application of XAI techniques, this study seeks to achieve a series of objectives aimed at enhancing the fairness and transparency of AI hiring systems.

A. BACKGROUND

The business hiring procedure has transformed due to the fast development of AI technology. AI-powered solutions are being used more and more to improve decision-making,

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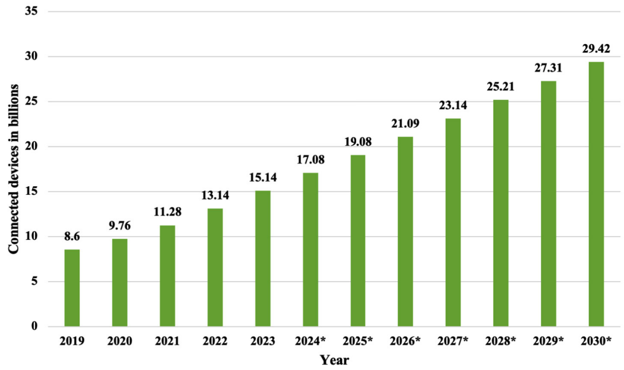

FIGURE 1. **Institutional myths, organizational practices, and reproduction inequalities. Note: Adapted from the organizational reproduction of inequality [3].**

expedite hiring procedures, and maximize talent acquisition. However, the widespread use of AI models in hiring processes caused worries about possible biases that can worsen already-existing employment disparities or even create new ones. It is now imperative to use XAI approaches to mitigate bias in AI models. This study examines the critical role that XAI plays in mitigating bias in AI models, especially when it comes to the recruiting practices used by the sector. The obvious existence of bias in hiring processes emphasizes the necessity of addressing bias in AI models. Alarming data from recent research have shown that automated employment systems exhibit bias against racial minorities, gender discrepancies, and certain socioeconomic backgrounds more than others [1], [2].

These biases not only support inequality but also work against the core values of meritocracy and fairness in the job market [3], [4]. Research has found that automated hiring systems can exhibit biases that favor certain demographic groups while disadvantaging others [1]. These biases can manifest in various forms, including racial, gender, and socioeconomic disparities. Racial minorities, for example, may face discrimination in AI-based hiring, leading to disparities in employment opportunities and perpetuating racial inequality in the job market [1]. Gender biases have also been identified in AI-powered hiring processes, with women often being under-represented or unfairly evaluated, hindering gender equality in the workforce [2]. Additionally, AI-driven recruitment systems have been shown to favor candidates from privileged socioeconomic backgrounds, further exacerbating societal divides [3]. These biases challenge the notion of meritocracy, where candidates should be evaluated solely based on their qualifications and abilities, and are at odds with principles of fairness and equal opportunity [4].

Fig. 1 illustrates the connections between myths, practices, and inequalities in AI-driven hiring processes [3]. To address these alarming biases, the incorporation of XAI approaches has become an important strategy. XAI offers transparency and interpretability in AI models, allowing organizations to understand and rectify the biases that may exist within their recruitment processes [5].

A generic Machine Learning (ML) approach involves teaching a computer to learn patterns from data without explicitly programming it. This method is good at making predictions but might not always explain how or why it reached a specific decision. On the other hand, XAI focuses on creating models that not only make accurate predictions but also provide understandable explanations for their decisions. XAI aims to make the black box of traditional ML models more transparent by providing insights into the factors influencing predictions [6]. By using XAI, this paper better understands and trusts the decisions made by AI systems, making them more accountable and interpretable for users. By employing XAI, companies can ensure fairness and equity in their hiring practices, ultimately promoting diversity and inclusion. XAI provides a level of transparency that allows organizations to comprehend how AI models arrive at their decisions. This transparency is essential for identifying and rectifying biases [5]. XAI tools can help organizations detect bias in their AI models by revealing the specific features or data points that contribute to biased outcomes. This information is critical for bias mitigation strategies. With XAI, AI models can provide explanations for their decision-making processes, offering insights into why a particular candidate was selected or rejected [6]. This transparency fosters trust and accountability. Employing XAI in hiring practices helps companies adhere to legal and ethical standards, reducing the risk of lawsuits and damage to reputation [7]. The integration of AI technology into the hiring process has introduced numerous benefits, but it has also unveiled significant biases that threaten the principles of fairness and equity in employment. To address these pressing concerns, the adoption of XAI has become essential. XAI empowers organizations to understand, detect, and mitigate biases within their AI models. Its transparency and interpretability provide valuable insights into the decision-making processes, allowing for fairer and more equitable recruitment practices.

B. PROBLEM STATEMENT

AI models are being used more and more by the industry to help with applicant selection as part of its efforts to optimize the recruiting process. Even though these models are supposed to be efficient and scalable, there is growing worry that they can unintentionally reinforce pre-existing biases or even make them worse, which would result in unfair application treatment and the continuation of structural inequalities. There are several potential origins of this bias, including model design, historical data, and the interpretability of AI conclusions. The problem gets worse using 'black box' AI models, which provide little to no visibility into the decision-making process and make it challenging to recognize, comprehend, and address biases [9]. Stakeholders cannot fairly evaluate the recruiting process or ensure the best individuals are selected regardless of their background if there is a lack of openness. It is not only a technological

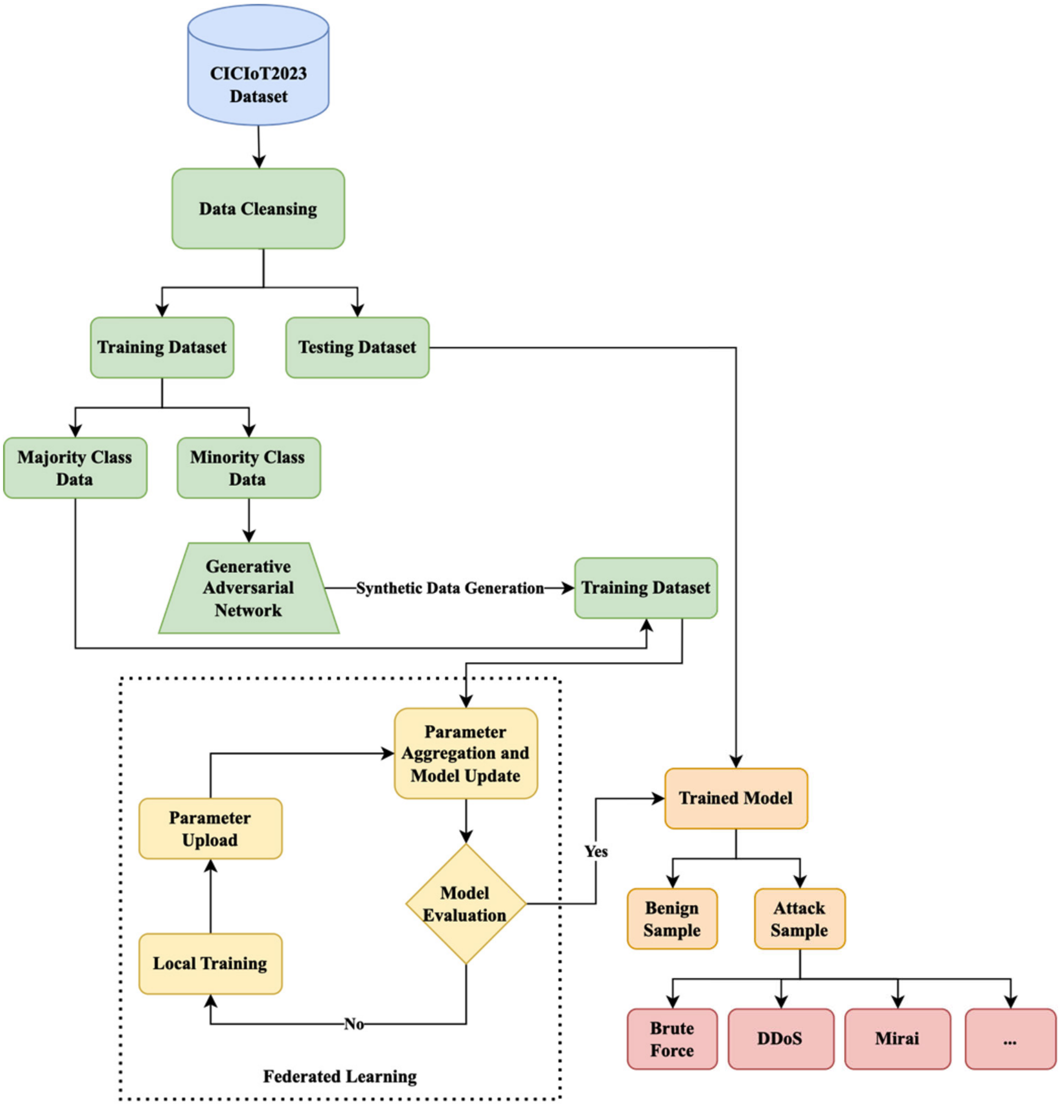

FIGURE 2. Generic approach compared to XAI approach [8]. Note: Adapted from A review of trustworthy and eXplainable Artificial Intelligence (XAI) [8].

problem but also an ethical need to preserve the principles of inclusiveness and fairness in the workplace that transparent, equitable, and responsible AI be used in hiring. By shedding light on the decision-making process of AI models and the characteristics that have the greatest impact on them, as shown in Fig. 2 XAI proves to be a crucial response to this issue.

C. AIMS AND OBJECTIVES

The purpose of this study is to demonstrate how biases in hiring-related AI models can be reduced by integrating XAI. By leveraging XAI approaches, 'black box' models can be transformed into 'glass box' systems, enabling us to:

- Monitor, understand, and modify the recruiting process.
- Ensure a more just and equitable hiring process that aligns with the industry's commitment to diversity and inclusion.

This research aims to develop a final glass box model, or a model without bias, using a systematic approach that includes:

- Modifying features to eliminate biases.
- Continuously evaluating model performance.
- Conducting rigorous explanatory analysis.

The primary goal is to create a more equitable and transparent AI system while ensuring that all parties involved in the hiring process can understand and defend the system's decisions. The research objectives include:

- Assessing the performance and potential biases in existing 'black box' AI hiring models (Random Forest, eXtreme Gradient Boosting (XGBoost), and Light Gradient Boosting (LightGBM)).
- Applying XAI methods to these models to expose and understand their decision-making processes.
- Identifying and altering features that contribute to bias.

- Iteratively tuning the models to balance high performance and reduced bias.
- Developing and refining glass box models free from bias with clear explanations for their decisions.
- Empirically demonstrating the advantages of using XAI in developing fair and transparent AI hiring systems through performance and fairness metrics comparison before and after the application of XAI methods.

D. RESEARCH CONTRIBUTION

- Through empirical demonstration, this research contributes to the field of ethical AI by showing how XAI may be used to detect and reduce biases in AI-driven hiring procedures.
- The particular framework converts opaque AI models into clear, comprehensible, and equitable decision-making mechanisms by using XAI approaches.
- This research offers a reproducible methodology for implementing AI systems free from bias in hiring, which will encourage inclusion and diversity in the workplace.
- It also offers practical insights for business practitioners.

II. LITERATURE REVIEW

The advent of AI technologies has revolutionized many sectors, including Human Resource Management (HRM). One of the most significant impacts has been observed in the hiring process, where these technologies promise to streamline operations, enhance decision-making, and potentially reduce human biases. However, alongside the benefits, there arises a critical concern regarding the biases inherent within AI algorithms themselves. This section discusses the dual-edged nature of AI in hiring, examining both its transformative potential and the challenges it poses, particularly focusing on algorithmic bias. By exploring a wide array of studies, this section aims to provide a comprehensive overview of the current landscape of AI applications in recruitment, the principles underpinning these technologies, and the ethical implications of their use. Through examining various studies, this section highlights the advancements, accuracy, and challenges associated with AI-driven recruitment processes, by providing a comprehensive overview of the theoretical foundations and key concepts surrounding Artificial Intelligence (AI) usage in recruitment.

A. OVERVIEW OF THE AI IN HIRING AND BIAS IN AI

AI is being increasingly used in hiring processes to find and assess candidates for open positions. It can supplement or replace traditional recruiting methods, such as screening applications and conducting interviews [10]. AI-based recruitment tools are changing the way recruitment processes are conducted, and applicants generally perceive AI technology positively in hiring processes. The advantages of AI in recruitment include reduced response time, while the drawbacks include the lack of nuance in human judgment, low accuracy and reliability, and immature technology [11]. AI-enabled interview analysis can provide deeper insights into candidate suitability, skills, and competencies, leading to more informed and objective decision-making in HRM [11]. But the use of AI in hiring processes also carries risks, particularly in terms of potential discrimination [12]. Businesses that want to use AI in their recruiting process need to exercise caution and take the appropriate precautions to reduce the possibility of prejudice. HR procedures and employee experiences have been significantly impacted by the development of AI in HR. AI-driven solutions are being more and more integrated by organizations into HR operations such as hiring, training, performance management, and employee engagement [13]. AI has the potential to revolutionize HR processes by predicting, recommending, and communicating based on data, leading to increased efficiency and scalability. The implementation of AI in HR functions can face challenges initially, but it is considered a valuable tool once successfully implemented [13]. The following are some examples of AI-based recruiting tactics: social media screening, chatbots, video interviews, resume screening, applicant matching, gamification, virtual reality evaluations, and chatbots [14]. These strategies offer significant benefits such as improved efficiency, cost savings, and better-quality hires. AI technology has helped improve HR activities such as recruitment, training, and career planning, reducing human effort and improving efficiency [13]. The efficiency and efficacy of HR procedures might be raised by implementing AI in HR management, improving company performance, and enhancing competitiveness [15]. Businesses should concentrate on Information Technology (IT) infrastructure, HR training, and fostering an AI-friendly corporate culture to enhance AI capabilities in HR management [15]. However, there are also moral and legal issues with AI usage in hiring, such as algorithmic prejudice and discrimination [11], [16], [17], [18], [19].

B. PRINCIPLES OF XAI

In discussing the principles of XAI, one can start by emphasizing the importance of transparency. XAI aims to make the workings of AI models understandable to humans [20]. This is crucial because it allows users and stakeholders to gain insights into how decisions are made. By doing so, trust is built between AI systems and their users. Transparency also enables accountability. When AI systems are used in critical areas, like hiring or healthcare, understanding their decision-making process is essential [21], [22]. This way, if an AI system makes an error, it can be traced back and corrected. Another key principle of XAI is fairness [20]. XAI seeks to ensure that AI decisions are made without bias towards any group or individual. This is important in promoting equity and preventing discrimination. By making AI systems explainable, it becomes easier to identify and eliminate biases that may exist in the data or the model itself [23]. Interpretability is also central to XAI [20]. It refers to the ability to present the workings of an AI model in a way that is understandable to humans. This

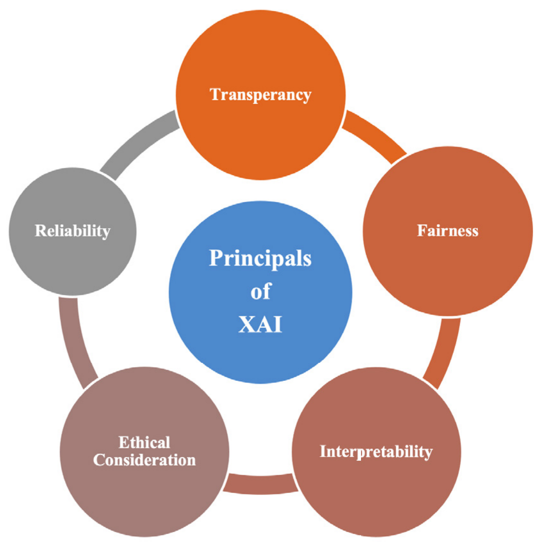

FIGURE 3. Principles of XAI – transparency, fairness, interpretability, reliability, and ethical considerations.

means that not only can experts comprehend how the AI arrived at a decision, but laypersons can too. This widespread understanding is crucial for the adoption and acceptance of AI technologies across different sectors. Moreover, the principle of reliability ties into the core of XAI [20]. An XAI system can consistently produce accurate and reliable results. Knowing how the system works increases confidence in its reliability, as stakeholders can see that it follows logical steps to reach its conclusions. Lastly, ethical considerations are fundamental to XAI [24]. The development and deployment of AI systems must adhere to ethical standards, ensuring that these technologies benefit society. Explainability aids in this by ensuring that AI systems operations align with societal values and norms, promoting a positive impact [24]. The principles of XAI revolve around making AI systems more transparent, fair, interpretable, reliable, and ethically sound, as shown in Fig. 3. These principles are intertwined, each supporting the other to create AI systems technically advanced, responsible, and trustworthy.

C. RELATED WORKS

The integration of AI algorithms in the recruitment process has seen a significant evolution over recent years. This section reviews recent studies directly relevant to the research, focusing on the purpose, themes, results, and identified limitations or biases within various studies to illustrate how our work addresses existing gaps in the literature on AI-driven hiring systems.

1) KEY STUDIES AND THEIR FINDINGS

The researcher proposed an AI-based system for the preliminary rounds of recruitment, emphasizing automation and AI assessments. Their work showcases the reduction in human intervention but falls short in discussing the model's accuracy or addressing potential biases, leaving room for questioning the fairness and reliability of the system [25]. The researchers developed an intelligent predictive model for effective candidate screening in IT firms using K-Nearest Neighbors (KNN), KMeans, and CART, achieving an impressive accuracy of 96.96%. However, the research may be susceptible to bias, though the authors have not acknowledged this potential limitation within the paper. Consequently, readers should critically assess the findings while considering the possibility of inherent biases influencing the results [26]. The researchers investigated job-seekers' perceptions of AI selection in job advertisements, finding negative effects on pre-process perceptions, especially when interviews are conducted by AI, highlighting the psychological impacts of AI automation on applicants [27]. The researchers discussed the challenges and outcomes of AI in hiring decision-making, noting its adoption primarily by high-tech or large companies. The study identifies a lack of detailed evaluation and the persistence of space for human bias, underscoring the need for a more comprehensive assessment [28]. The researchers through speculative design, engaged stakeholders in deliberations on the future of AI in recruitment. Despite facilitating meaningful conversations, the study lacks a detailed exploration of stakeholder views, and the study may exhibit biases; however, the author neglected to acknowledge them within the paper [29].

2) LIMITATIONS AND BIASES IN EXISTING RESEARCH

The researchers raised ethical concerns regarding AI in candidate selection, particularly for cyber vetting purposes. The potential for AI biases to disqualify competent candidates underscores the ethical dilemmas inherent in these technologies [30]. The researchers suggested the use of ML to enhance the multi-level fit personnel selection process and provide the 'Quality of Hire' idea. The absence of reported accuracy and the difficulty in interpretation by hiring managers present challenges in practical application [31]. The researchers aimed to facilitate the employee selection process using a Bi-Directional Encoder Representation Transformer (BERT), Latent Semantic Analysis (LSA), and Support Vector Machine (SVM), achieving an accuracy of 81% with SVM. The research may harbor biases that were not explicitly mentioned by the author within the paper, potentially impacting the study's objectivity and reliability. Acknowledging and mitigating such biases is crucial for ensuring the credibility and validity of the research findings [32].

The researchers evaluated the effectiveness of ML algorithms in hiring decisions, reporting an accuracy of 73.79% with Decision Tree and highlighting a significant bias indicated by poor recall value. This study provides valuable insights into the limitations and challenges of applying ML in hiring [33]. The researcher examined innovations, patterns, and best practices in HRM, encompassing performance management and AI and ML integration. The lack of specific details on biases within models points to an area that requires further exploration for a holistic understanding [34]. The researchers focused on analyzing employee turnover using ML algorithms, achieving an 88.2% accuracy with Adaptive Boosting (AdaBoost). The author suggests a potential oversight or omission in mentioning methodological limitations. Without explicit acknowledgment, readers may need to critically evaluate the study's findings in light of potential biases that could influence its conclusions [35]. The researchers aimed to enhance decision-making processes in HR through improved accuracy in analyzing and predicting employee attrition, using a Decision Tree, Random Forest, and Binary Logistic Regression, achieving an accuracy of 87.44% with Logistic Regression. The acknowledgment of potential biases without detailed discussion suggests an area ripe for further investigation [36]. The researchers critically analyze claims by recruitment of AI companies regarding their ability to eliminate bias, finding that such claims are often misleading, reflecting a misunderstanding of the underlying issues and power dynamics. They highlight the risk of outsourcing diversity efforts to AI could exacerbate rather than alleviate inequality [37]. The researcher delves into the themes of exclusion and inclusion within AI-supported hiring, emphasizing the need for algorithmic audits. The study points out that AI can perpetuate existing biases, underlining a significant gap in research regarding human-machine interactions in decision-making processes [38]. The researchers enhance the conversation by putting out a methodology to evaluate the parts played by developers and HR managers in the development of discriminatory AI hiring practices. Their work underscores the importance of recognizing cognitive biases, such as the 'similar-to-me' and stereotype biases, as critical to developing unbiased AI. However, they note a limitation in their research: a lack of concrete guidelines for removing bias from these systems [39]. The researchers explore a more technical solution with their development of the SensitiveNets algorithm, aimed at excluding sensitive information from AI decision-making processes to foster fairness. This approach demonstrates the potential to generate fairer AI-based recruitment tools by focusing on non-textual data. Yet, the study's limitation lies in its exclusive analysis of image data, neglecting the impact of textual information in perpetuating bias [40].

3) GAPS IN THE LITERATURE

While the reviewed literature demonstrates significant advancements in applying AI algorithms in the recruitment process, a recurring theme is the lack of detailed exploration into biases within these models. A recurring gap across these studies is the lack of thorough exploration into potential biases inherent in the ML algorithms utilized. While many studies report impressive accuracy rates, they failed to address the issue of bias within these models. Furthermore, despite the acknowledgment of bias as a concern by some researchers, the detailed discussions on how to address these biases remain largely absent. This gap indicates the need for research that not only assesses model performance but also critically evaluates and addresses biases to ensure fairness and transparency in AI-driven hiring systems.

4) CONTRIBUTION OF THIS STUDY

Our research aims to fill this gap by systematically evaluating the performance and biases of Random Forest, XGBoost, and LightGBM algorithms. By employing XAI methods, this study to reveal the decision-making processes of these models and identify features contributing to bias. Subsequently, through iterative tuning and refinement, this paper develops glass box models that are not only high-performing but also free from biases, providing clear explanations for their decisions. Through empirical demonstration, this study shows the advantages of integrating XAI in the development of fair and transparent AI hiring systems, thereby contributing to a more ethical and equitable recruitment process. By addressing the research gap identified in previous works, our study aims to provide practical insights and solutions for creating more inclusive and unbiased AI-driven hiring practices. Table 1 summarizes key related works, emphasising their methods/used algorithms, focus, bias handling, and limitations, showing how our study offers bias mitigation and explainability in AI hiring.

III. MATERIALS AND METHODOLOGY

As shown in Fig. 4 the proposed methodology for the study and address biases in AI-driven hiring models through the integration of XAI techniques is outlined. The methodology encompasses data collection, model development, XAI integration, and performance evaluation stages. By systematically delineating each step, this section provides a comprehensive framework for achieving transparency, fairness, and efficacy in AI-driven hiring processes. The development of an unbiased AI model for hiring involves a series of carefully planned and executed steps designed to create a system that is both effective in identifying suitable candidates and equitable in its operation. The steps are as follows:

The development of a fair and transparent AI hiring model begins with assembling a diverse dataset to ensure the data is representative of the population, thus avoiding biases from unrepresentative data. The next step involves training a baseline 'black box' AI model using standard algorithms to establish initial performance metrics and identify potential biases. This baseline model is then analyzed for biases, and features contributing to bias are altered or removed to mitigate their impact on decision-making. Continuous monitoring and tuning of the model, balancing accuracy with the need to reduce bias. XAI techniques are integrated to make the decision-making process transparent and understandable to humans, crucial for evaluating the fairness of decisions. As a result, the model is transformed from a 'black box' to a 'glass box' model, where decisions can be fully explained and justified. The refined model undergoes

TABLE 1. Comparative analysis of key related works and the present study.

Study	Methods	Focus Area	Accuracy	Bias Consideration	Explainability	Limitations	Contribution of Our Study
[26]	KNN, KMeans, CART	Candidate screening in IT firms	96.96%	Not acknowledged	No	Potential bias unaddressed	Explicit bias identification and reduction.
[32]	BERT, LSA, SVM	Employee selection	81%	Mentioned but not detailed	No	Biases not explicitly mitigated	Bias reduction and transparent models.
[36]	Decision Tree, Random Forest, Logistic Regression	Employee attrition prediction	87.44%	Potential bias acknowledged	No	Limited bias discussion	Explicit bias handling and model transparency.
[39]	Methodology to evaluate cognitive biases	Developer & HR roles in AI hiring	N/A	Biases acknowledged, no removal guidelines	No	Lack of concrete bias removal strategies	Concrete bias mitigation strategy with XAI.
[40]	SensitiveNets algorithm	Fairness via sensitive info exclusion	N/A	Partial (image data only)	No	Textual data bias unaddressed	Broader bias evaluation including textual data.
Our Study	Random Forest, XGBoost, LightGBM with XAI	Bias mitigation, explainability	Competitive	Comprehensive bias evaluation	Yes	N/A	High-performance, unbiased, interpretable AI.

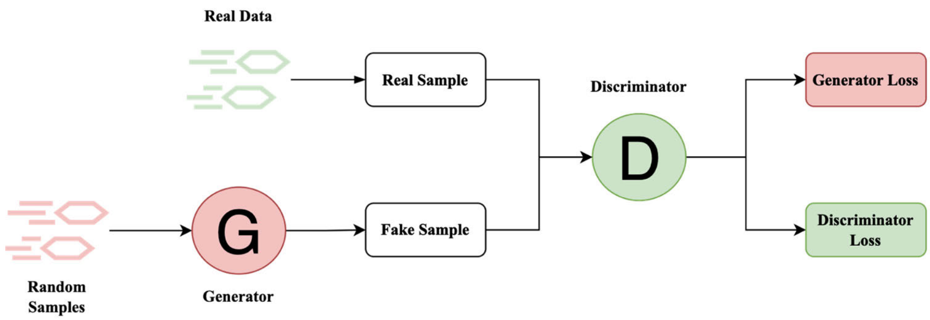

FIGURE 4. Proposed methodology of the unbiased AI model.

rigorous testing to ensure it performs well according to established metrics and that bias has been effectively reduced. Finally, the performance and fairness of the 'glass box' model are compared to the original 'black box' model, demonstrating improvements in fairness and transparency without compromising the effectiveness of the hiring process.

A. HIRING DATASET COLLECTION

To ensure the AI model is trained on high-quality data that is reflective of the real-world hiring environment. The dataset should include features that are relevant to the hiring decisions, such as qualifications, experience, skills, and any other attributes that are essential for the job roles in question. The dataset must include a wide range of examples across different job positions, industries, and company sizes to ensure the model can be generalized across various hiring contexts. The dataset should include a proportional representation of different demographics, including gender, age, ethnicity, and socioeconomic backgrounds, reflecting the diversity in the workforce. For this study, two datasets will be used, namely, Utrecht Recruitment Dataset and Hybrid Hiring Dataset. The datasets were collected from Kaggle and Microsoft websites, respectively [41], [42]. The Utrecht Recruitment dataset encompasses the hiring outcomes from four distinct firms, encompassing a pool of 500 applicants. The dataset provides a basic profile for each individual, detailing attributes such as gender, age, and sporting interests, alongside several performance indicators. Included as well is the ultimate hiring verdict for each candidate. The hiring choices made by these organizations exhibit varying degrees of bias, presenting a fertile ground for scholarly scrutiny. Researchers have the opportunity to dissect bias across a spectrum of dimensions, including gender, nationality, age, and athletic involvement. To enrich the dataset's utility for bias analysis, additional context on company cultures and hiring protocols is also provided, allowing for a more nuanced understanding of how and why certain biases might manifest in these decisions [41]. Microsoft has launched an extensive dataset known as Hybrid Hiring, which serves as a substantial resource for examining the interplay between human and AI decision-making in real-world hiring scenarios. This dataset, unprecedented in its release, consists of 38,400 human evaluations across 9,600 distinct predictive tasks, distributed over seven experimental conditions. Hybrid Hiring is designed to provide a comprehensive look at how people make decisions when they are informed by ML predictions, offering invaluable insights into the integration of AI in human decision processes [42].

B. MODEL DEVELOPMENT

1) TRAINING THE BLACKBOX AI MODEL

When embarking on the development of an AI model for hiring, the choice of the algorithm is critical. This choice can be influenced by the type of data available, the complexity required, and the performance metrics this paper to optimize. This study will use the Random Forest algorithm for decision-making. Several decision trees are built during the training phase of Random Forest, an ensemble learning technique for classification and regression, and the class that results is the mean prediction (regression) or mode of the classes (classification) of the individual trees [42]. Random Forest is correct for decision trees' habit of overfitting to their training set. The decision trees that make up the 'forest' are often trained using the 'bagging' technique. The fundamental concept behind bagging is to combine multiple models (in this case, trees) to improve the overall result [43]. Random Forest creates a lot of individual decision trees using a technique called bagging. It starts by randomly selecting samples from the dataset with replacements (meaning the same sample can be chosen more than once) to create multiple different subsets of the data [42]. For each new subset, a decision tree is created. During the construction of these trees, only a random subset of the features is considered for splitting at each node, which is a key aspect that differentiates Random Forest from Bagging. This randomness ensures that the trees are different and can capture various patterns in the data [43]. For a classification problem, each tree in the forest votes for a class, and the class with the most votes becomes the model's prediction [44]. For regression problems, the average of the outputs of all trees is taken as the prediction [43]. The individual trees' predictions are then aggregated to produce a single result. The aggregation process averages the predictions for regression problems and uses majority voting for classification problems [42], [43], [44]. Random Forests tend to be very accurate classifiers and regressors. The ensemble of diverse trees mitigates the errors from individual trees, often leading to better performance than single decision trees [44]. It is less likely to overfit than a single decision tree, especially on large datasets. Random Forest has the innate ability to rank features by their ability to improve the purity of the node, which can be used for feature selection. It can maintain accuracy even when a large proportion of the data is missing [42]. Random Forest algorithms are relatively easy to use as they require very few tuning parameters and can be used as a starting point for more complex algorithms.

2) INITIAL TRAINING AND PERFORMANCE EVALUATION

With the algorithm selected and the model configured, initial training can commence. The model is trained on the training set and its performance is assessed on the validation set once the dataset is split into training and validation sets. The following are included in the first training phase [45], [46]. The model is trained on the dataset, consisting of features from the hiring dataset and target variables (e.g., successful, and unsuccessful hires). Performance is evaluated using metrics, such as F1-score, recall, precision, and accuracy, along with fairness metrics like demographic parity or equalized odds to assess bias. Bias assessment is conducted by analyzing predictions across different demographic groups to ensure fairness. The model's predictions are compared against actual outcomes in the validation set to determine performance, and errors are analyzed to understand their cause. This phase results in a trained 'black box' AI model, setting the stage for subsequent steps to make its decisions explainable and mitigate biases through XAI techniques.

C. XAI INTEGRATION

The Random Forest classifier, while robust and powerful, is often considered a 'black box' model due to its complexity involving numerous decision trees. To enhance its explainability, this paper implements SHapley Additive exPlanations (SHAP) as an XAI technique which allows us to interpret the model's decision-making process.

1) SHAPLEY ADDITIVE EXPLANATIONS (SHAP)

SHAP is a powerful and widely used technique in the field of XAI. It is designed to provide interpretable insights into the output of ML models by attributing the contribution of each feature to every prediction [47]. The visual representation assists in discerning the predominant attributes and their respective contributions, gauged by the probability value. The SHAP model elucidates the impact of a feature on the target variable SHAP values are based on cooperative game theory concepts, specifically the Shapley values, which allocate the 'worth' of each player in a cooperative game. SHAP values are grounded in the Shapley values, which were originally introduced in cooperative game theory to fairly distribute the payoff among participants based on their contributions [48]. In the context of ML, features are treated as 'players' in a cooperative game, and the Shapley values determine how to distribute the model's prediction among these features. SHAP values can be visualized using SHAP summary plots, which display the average impact of each feature across the dataset [47]. Individual SHAP value plots show how each feature contributes to a specific prediction. These visualizations help in identifying which features push the model's prediction higher or lower for a given instance [48].

2) PURPOSE AND SIGNIFICANCE OF USING XAI IN HIRING PROCESS

The utilization of XAI in hiring processes addresses critical concerns surrounding fairness, transparency, and bias in AI-assisted recruitment. Driven by the urgent need to ensure fairness and prevent discrimination, XAI provides clarity on how AI models make decisions, which is typically opaque in traditional 'black box' models. This transparency is crucial for several reasons. Firstly, it enables accountability by allowing the tracing of decision-making paths and holding systems accountable for their decisions, especially in hiring where decisions significantly impact individuals' careers. Secondly, XAI helps in bias mitigation by identifying and addressing biases that may exist due to historical data or model construction, promoting equity. Thirdly, it ensures compliance with regulations that mandate explainability and non-discrimination in AI systems used in critical areas like hiring. Additionally, XAI helps in building trust among candidates and employers, as they are more likely to trust AI systems when they can understand how and why certain decisions are made, which is essential for the adoption of AI in hiring. Finally, XAI provides insights into the continuous improvement of AI models, ensuring they remain relevant, fair, and effective as societal norms and legal requirements evolve [49].

3) MONITORING EXPLANATIONS FOR INDICATIONS OF VIAS

Monitoring the explanations produced by XAI techniques allows for vigilant checking of potential biases, achieved through several methods. Firstly, evaluating disparate impact involves using insights from explainability techniques to look for patterns indicating a disproportionate impact on different demographic groups. If certain features disproportionately affect minority groups, it could signify bias. Secondly, consistency checks compare explanations across similar instances to identify any inconsistencies, which can indicate underlying bias in the model. Thirdly, counterfactual explanations are generated to understand the minimal changes needed to alter a prediction. Observing which features often appear in counterfactuals can highlight potential biases, especially concerning protected attributes. Additionally, alongside explainability, fairness metrics are computed and monitored, such as equal opportunity and demographic parity, to quantitatively assess the model's fairness. The continuous monitoring of these explanations and metrics aids in identifying and addressing biases, ensuring that the Random Forest model adheres to ethical standards and promotes fairness and transparency in AI-based decision-making processes [49], [50].

D. FINAL MODEL DEVELOPMENT

The transition from a 'black box' to a 'glass box' model in the context of AI for hiring involves transforming a model whose internal workings are opaque into one that is transparent and explainable. This section focuses on using a Random Forest classifier, which is inherently more interpretable than many other ML models, yet still requires explicit steps to ensure transparency and mitigate bias. This paper conducts a thorough analysis of the feature set to identify any variables that may introduce bias, such as those that correlate closely with protected attributes like race, gender, or age. This paper implements pre-processing techniques to ensure that the training data does not contain bias. This could include re-sampling methods or re-weighting instances to balance the representation of different groups. This paper also applies algorithmic fairness techniques such as fairness constraints or regularization methods that penalize the model for biased predictions. Afterward, the Random Forest model will be trained with a modified objective function that includes fairness constraints. These constraints could aim to equalize prediction rates across different groups, also known as demographic parity, or to ensure that similar individuals have similar prediction outcomes, known as equality of opportunity. A validation set that is representative of the general population will be utilized to test the model for fairness. Employ fairness metrics such as equal opportunity, disparate impact, and statistical parity to assess whether the model's predictions are unbiased. A thorough error analysis will be conducted to understand the types of errors the model is making and whether they disproportionately affect one group over another. For the Random Forest model to be explainable and its decisions understandable. Model interpretation frameworks designed for ensemble methods, such as feature importance scores, which indicate how much each feature contributes to the overall prediction, will be utilized. Techniques like Partial Dependence Plots (PDPs) and Individual Conditional Expectation (ICE) plots will be applied to show the relationship between the features and the prediction outcome. Instance-level explanation tools like SHAP will be utilized to provide explanations for individual predictions, helping stakeholders understand why a particular decision was made. By following these detailed steps, the Random Forest classifier can be developed into a final 'glass box' model that is not only free of bias to the greatest extent possible but also provides clear and understandable decision-making processes to its users.

E. PERFORMANCE AND FAIRNESS EVALUATION

The final stage of the methodology focuses on evaluating the performance and fairness of the Random Forest classifier, which has been modified through the XAI framework to mitigate bias. This evaluation is two-fold: firstly, assessing the model's performance against established benchmarks, and secondly, comparing the performance of the original (pre-XAI) and final (post-XAI) models.

F. METHODOLOGY FOR TESTING AGAINST ESTABLISHED BENCHMARKS

To test the final model, this paper adopts a comprehensive benchmarking strategy that considers various performance metrics to evaluate the model's accuracy, fairness, and overall effectiveness in a hiring context. Accuracy metrics are evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score to assess the model's predictive performance [51]. These metrics serve as quantitative measures to gauge how well the model is performing. Accuracy calculates the ratio of correctly predicted instances to the total number of instances in the dataset. Precision measures the accuracy of positive predictions, indicating the proportion of true positives among all instances predicted as positive. Recall gauges the model's ability to correctly identify all relevant instances by calculating the ratio of true positives to the total number of actual positives. The F1-score, as the harmonic mean of precision and recall, offers a well-balanced evaluation [51]. To ensure the model's robustness, k-fold cross-validation is implemented. This technique involves partitioning the data into k subsets, training the model on k-1 subsets, and validating it on the remaining subset. This process is repeated k times, with each subset serving as the validation set once [52]. Additionally, the model is tested on an external benchmark dataset commonly used in the AI fairness literature. This approach allows for a comparison of the model's performance with other studies and ensures that the findings are consistent with broader research.

G. COMPARING THE PERFORMANCE OF BLACK AND WHITE BOX MODELS

The comparison between the black box and the final models will highlight the impact of the XAI interventions on both performance and fairness. Performance comparison involves assessing the accuracy metrics of both models to identify any changes in predictive performance. While a slight reduction in accuracy is often anticipated when adjustments are made to reduce bias, the goal is to minimize this trade-off. Fairness comparison focuses on improvements in fairness metrics, indicating a reduction in bias and demonstrating the effectiveness of the methodology in creating a more equitable AI model. Feature importance analysis, particularly with Random Forest, examines how the XAI-guided alterations have shifted the model's decision-making by comparing feature importance distributions from both models. Additionally, decision boundary analysis visualizes the decision boundaries of both models to understand how the modifications have affected the model's decision space, providing insights into the complexity and generalization capabilities of the models. The goal of this evaluation is to demonstrate that the final model, informed by XAI, performs comparably to the original model in terms of accuracy while significantly improving in terms of fairness.

IV. RESULT ANALYSIS AND DISCUSSION

A. EXPLORATORY DATA ANALYSIS (EDA)

The bar chart in Fig. 5 shows the distribution of degrees among applicants across all companies. Bachelor's degrees are the most common among applicants.

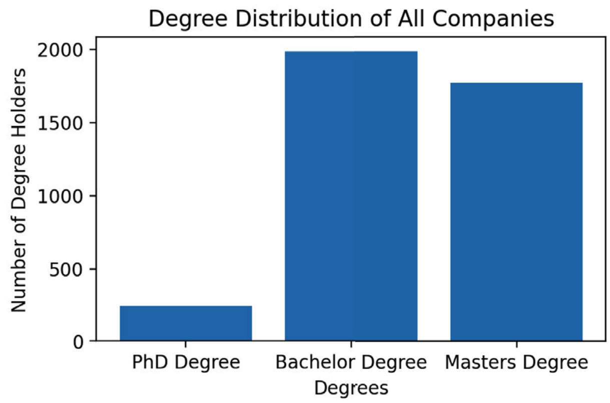

FIGURE 5. Distribution of educational degrees among applicants across all companies, showing a higher prevalence of Bachelor's and Master's degrees compared to PhDs.

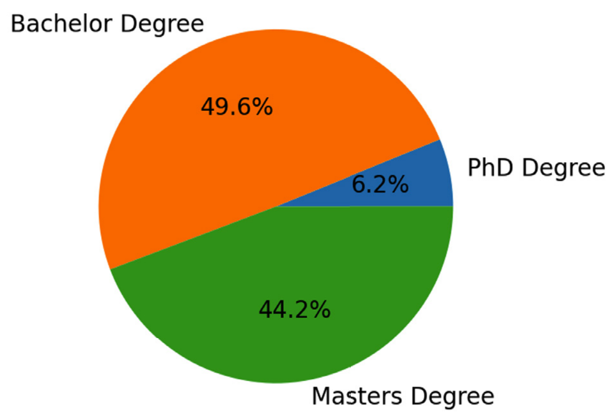

FIGURE 6. Percentage breakdown of degree types across all companies, highlighting the dominance of Bachelor's and Master's degrees.

The pie chart in Fig. 6 breaks down the degree distribution across all companies by percentage. Where the majority of applicants hold bachelor's and master's degrees. For individual companies (A, B, C, D), the degree distributions show a similar trend where bachelor's and master's degrees dominate in Figs. 7, 8, 9, and 10.

The pie chart in Fig. 11 of gender distribution indicates 53.2% of the applicants are male, 44.7% are female, and 2.1% are classified as other.

The histogram in Fig. 12 of age distribution shows a concentration of applicants between the ages of 24 and 30, with a peak at around 28 years old. This suggests that the majority of applicants are in their late twenties, which may impact age-related biases in hiring practices.

Each of these visualizations provides critical insights into the demographics and qualifications of the applicant pool, which are essential for understanding and potentially mitigating biases in AI-driven hiring models. This analysis can be further used to fine-tune the models for fairness and to ensure that hiring practices are inclusive and equitable across all demographic groups.

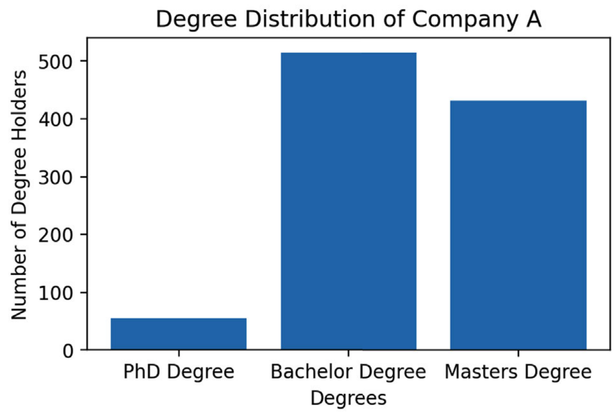

FIGURE 7. Degree distribution among applicants at Company A, with a notable preference for Bachelor's and Master's degrees over PhDs.

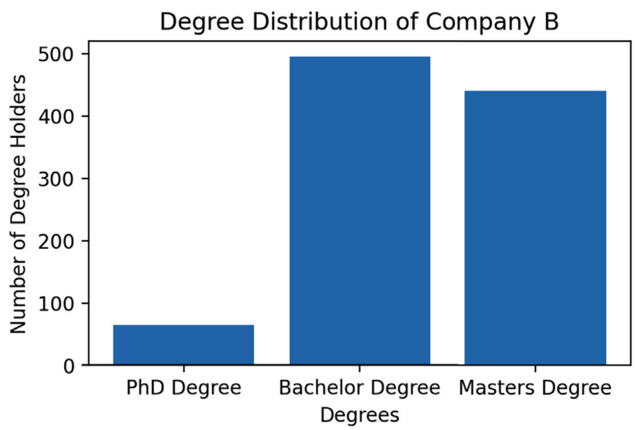

FIGURE 8. Educational qualification distribution at Company B, reflecting a similar trend of higher numbers of Bachelor's and Master's degree holders.

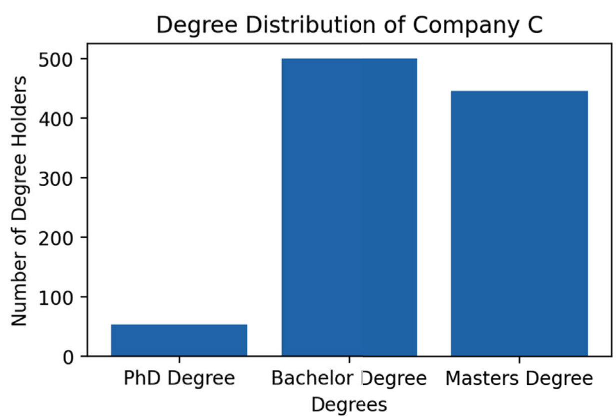

FIGURE 9. Comparison of degree holders at Company C, where Bachelor's and Master's degrees significantly outnumber PhDs.

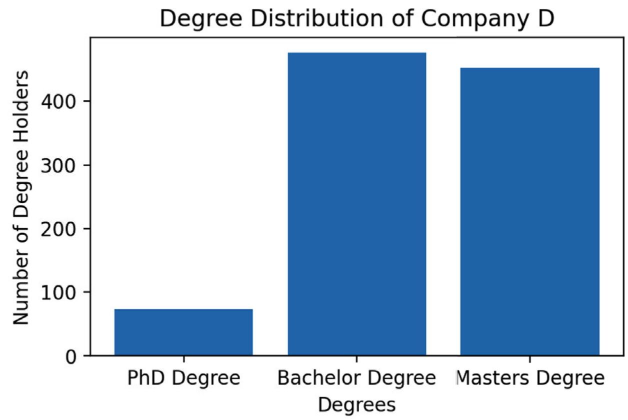

FIGURE 10. Degree distribution at Company D, indicating a consistent trend across companies with a lower representation of PhDs.

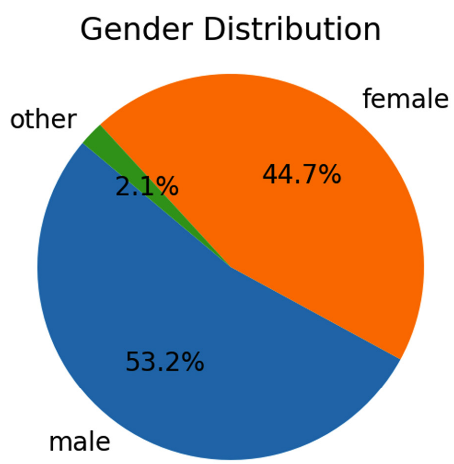

FIGURE 11. Gender distribution among applicants, with a majority of male applicants followed closely by female and other categories.

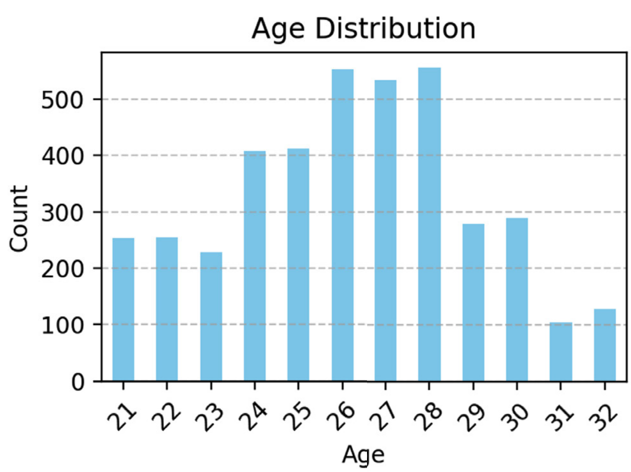

FIGURE 12. Histogram of applicant ages, with a concentration in the late twenties, peaking at 28 years, which could influence age dynamics in hiring decisions.

B. ANALYSIS OF BLACK BOX MODEL PERFORMANCE

The performance of the black box model, without any feature elimination, reveals significant insights about biases inherent in the model. With an accuracy of 0.7 and, a precision of 0.60, recall of 0.66, and an F1-score of 0.63, the model shows a reasonable imbalance between correctly predicting positive cases and handling the overall dataset, as presented in Fig. 13.

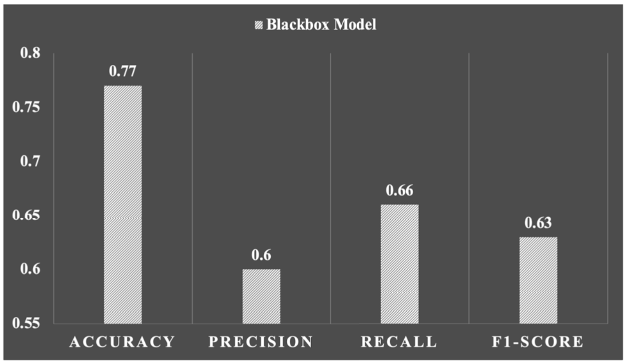

FIGURE 13. Figure summarizing the performance metrics of the black box AI model without feature elimination, detailing accuracy, precision, recall, and F1-score to evaluate model effectiveness and potential biases.

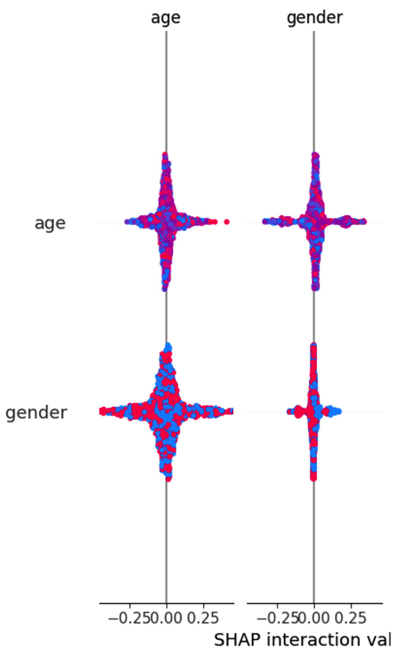

FIGURE 14. SHAP interaction plot showing the impact of age and gender on the model's decisions, illustrating how these features interact to influence hiring outcomes.

However, delving deeper into the model's behavior using correlation (in Fig. 14) and SHAP analysis (in Fig. 15) provides an understanding of how certain features, particularly 'gender', 'nationality', 'sport', and 'age', influence its decisions.

The SHAP interaction plot in Fig. 15 indicates that gender significantly impacts decision outcomes, with distinct SHAP values for males and females. This suggests that gender influences hiring decisions in a way that could perpetuate existing gender disparities in the workplace.

Similarly, the correlation heatmap shows a lack of a strong correlation between nationality and the hiring decision in Fig. 15, but its presence in the model can subtly influence outcomes based on geographical or cultural background, leading to potential nationality-based discrimination. The influence of sports choice on hiring decisions, as seen in the correlation heatmap in Fig. 15, raises concerns. Both 'ind_university_grade' and 'ind_exact_study' show strong correlations with each other and moderate to strong influences on hiring decisions. This indicates that academic performance and specific areas of study are significant determinants in the hiring process, which is justifiable given that they directly relate to job competencies. 'ind_programming_exp' and 'ind_international_exp' are heavily weighted in the model, suggesting that technical skills and global exposure are highly valued. These features have a positive impact on hiring decisions and align with the demand for skilled professionals in an increasingly globalized and technologically advanced job market. 'Entrepreneurial Experience' also shows a significant positive impact on decisions, highlighting the value placed on innovation and leadership skills within industries that favor entrepreneurial spirit. The 'black box' model's reliance on certain demographic features (like gender, nationality, and sport) that do not necessarily correlate with job performance indicates inherent biases that need addressing.

C. ANALYSIS OF WHITEBOX/GLASSBOX MODEL PERFORMANCE

1) ANALYSIS OF WHITEBOX/GLASSBOX MODEL PERFORMANCE (ELIMINATING "GENDER," "NATIONALITY," "SPORT")

The white box model, which was refined to eliminate biases associated with gender, nationality, and sport, demonstrates a shift towards more logical and job-relevant decision-making criteria. The bar chart in Fig. 16 displaying the performance metrics of the 'white box' model achieves an accuracy of 0.85, indicating that it can correctly identify the outcome in 85% of the cases, which is a robust performance indicator for most hiring contexts.

The recall rate of 0.74 shows that the model identifies 74% of all actual positive outcomes. The F1-score of 0.75 balances precision and recall, confirming that the model is effective in handling both the relevance and retrieval of potential hires.

The removal of features such as gender, nationality, and sport is significant. By eliminating this feature, the model focuses more on skill sets and qualifications, reducing gender bias. Removing this feature helps ensure that candidates are evaluated based on their competencies. While occasionally relevant to teamwork or leadership skills, sport is not universally essential for job performance. Its removal helps to streamline candidate evaluation processes towards more critical skills and experiences. The correlation heatmap in Fig. 17 and the SHAP interaction values in Fig. 18 further reveal the adjusted focus of the model. In the new model configuration, there is a clearer emphasis on professional and academic qualifications, such as university grades and specific experiences like programming and international exposure.

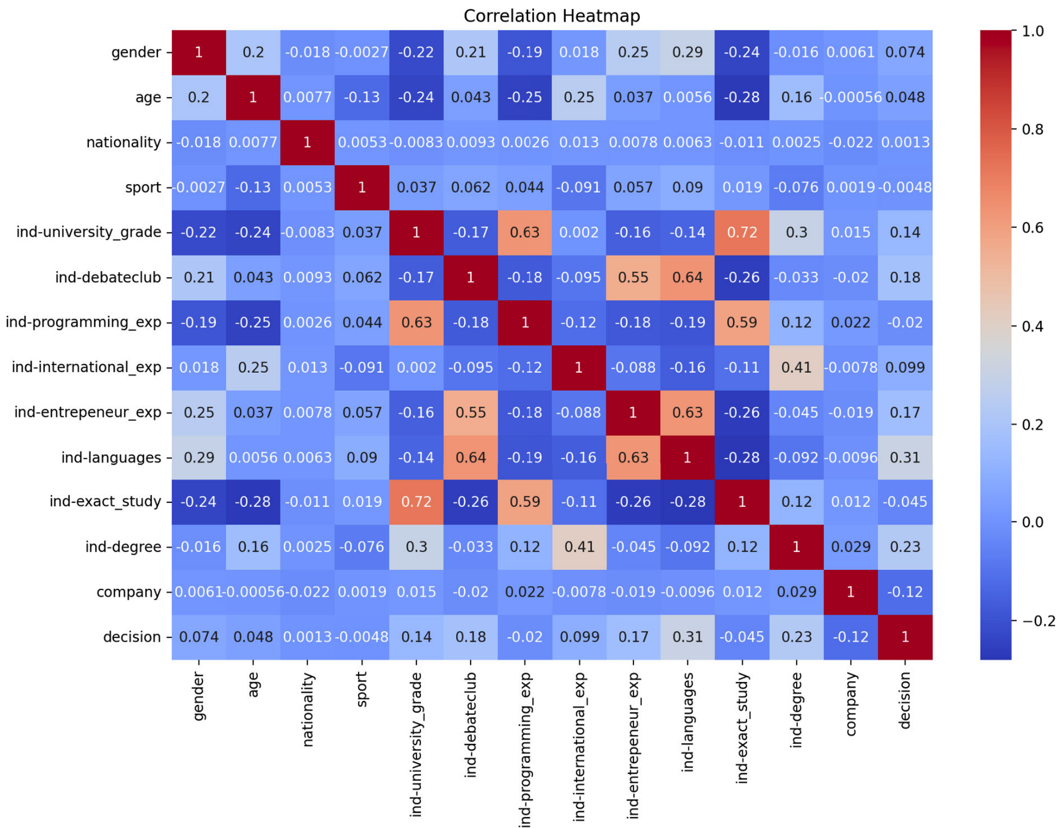

FIGURE 15. **Heatmap displaying the correlation coefficients between various features such as gender, age, nationality, and professional experiences, highlighting interdependencies and the potential for bias within the dataset.**

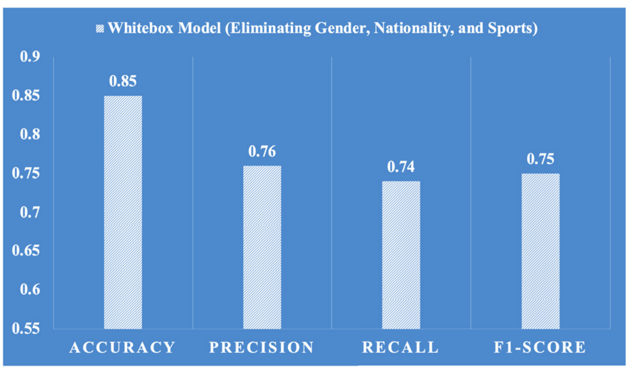

FIGURE 16. **Bar chart illustrating the performance metrics of the whitebox model after eliminating biases associated with gender, nationality, and sports. The chart shows accuracy, precision, recall, and F1-score, highlighting the model's efficiency in making it unbiased.**

These features have stronger and more direct correlations with hiring decisions, as they are more predictive of job performance. The interaction between age and university grades in the SHAP plot indicates that these factors now play a more significant role in the decision-making process. The model considers how these elements interact to impact performance predictions, aligning with logical hiring practices that value experience and education as indicators of potential employee success.

The 'white box' model performance, as illustrated by the high accuracy and balanced precision, recall, and F1-score, along with the removal of illogical biases, underscores its effectiveness and fairness. By focusing on job-relevant criteria and minimizing the impact of non-essential personal characteristics, the model upholds the principles of equitable AI in hiring practices.

2) ANALYSIS OF WHITE BOX/GLASS BOX MODEL PERFORMANCE (ELIMINATING "GENDER" ONLY)

In exploring the performance of the whitebox model specifically configured to eliminate 'gender' as a feature while retaining other attributes, such as 'nationality' and 'sport', the metrics and visualizations in Fig. 19 provide insights into the model's effectiveness and biases. The performance metrics of this white box model show an accuracy of 0.79, indicating a fair ability of the model to correctly predict outcomes. The precision of 0.64 suggests that the model has moderate reliability in predicting positive cases. The recall of 0.69 indicates that the model captures a good proportion of actual positive cases. The F1-score of 0.66 reflects an imbalance between precision and recall but indicates room for improvement. These metrics suggest that the model performs adequately but is not as effective as it could be, possibly due to retaining other bias-prone features.

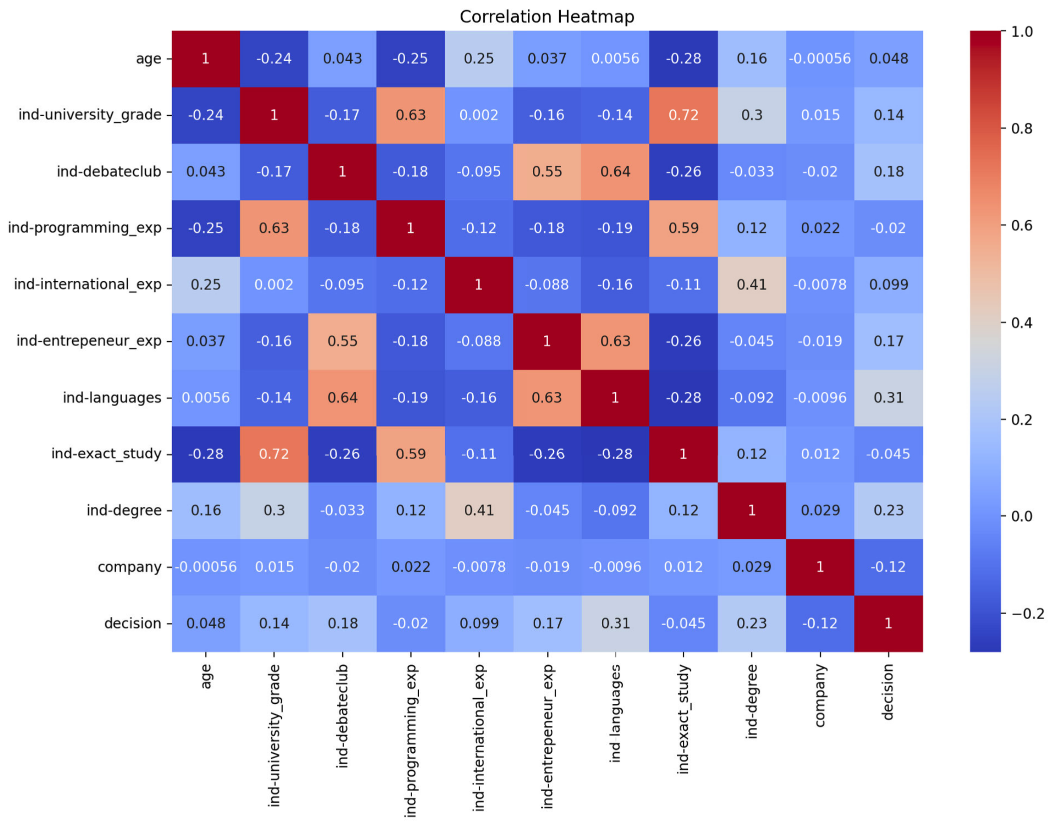

FIGURE 17. **Heatmap showing the correlation coefficients among key features in the white box model. This visual emphasizes the relationships and influence of academic and professional experience on hiring decisions after removing biased features.**

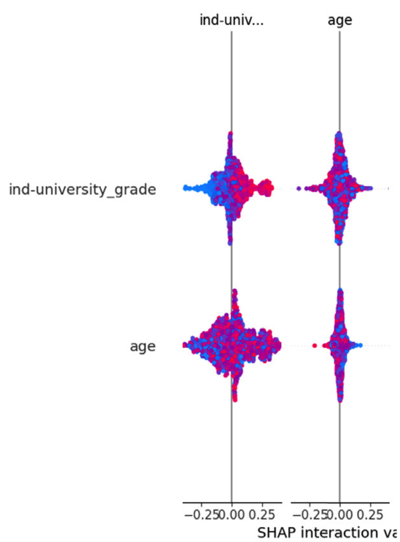

FIGURE 18. **SHAP interaction plot displaying the interaction effects between age and university grade on the model's predictions.**

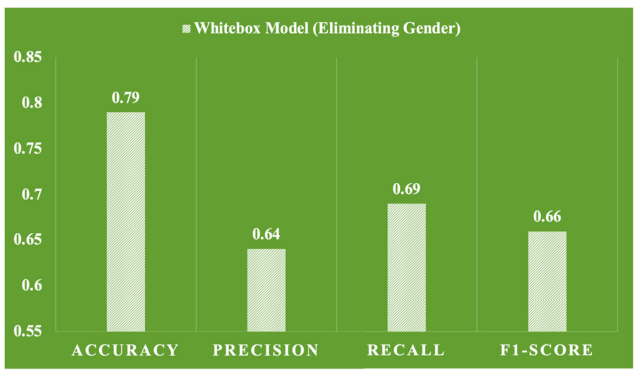

FIGURE 19. **Bar chart showing the performance metrics of the whitebox model after eliminating gender as a feature. Metrics include accuracy, precision, recall, and F1-score, indicating a moderate performance level.**

Despite eliminating gender, the model still includes nationality, which shows a strong bias in the correlation heatmap in Fig. 20 or the SHAP interaction plot in Figure 22.

The correlation heatmap in Fig. 20 Shows moderate to low correlations between sports and other professional skills. The parallel plot in Fig. 21 provides a detailed visualization of the flow and impact of each feature on hiring decisions. The strong visual paths from 'sport' indicate its significant weight in the decision process, which may overshadow more pertinent qualifications. SHAP interaction plot in Fig. 22 shows that while nationality and age interact to affect decisions the influence of nationality is relatively neutral compared to age.

The continued inclusion of 'nationality' and especially 'sport' as features introduces potential biases that may not be directly relevant to job capabilities. This analysis suggests

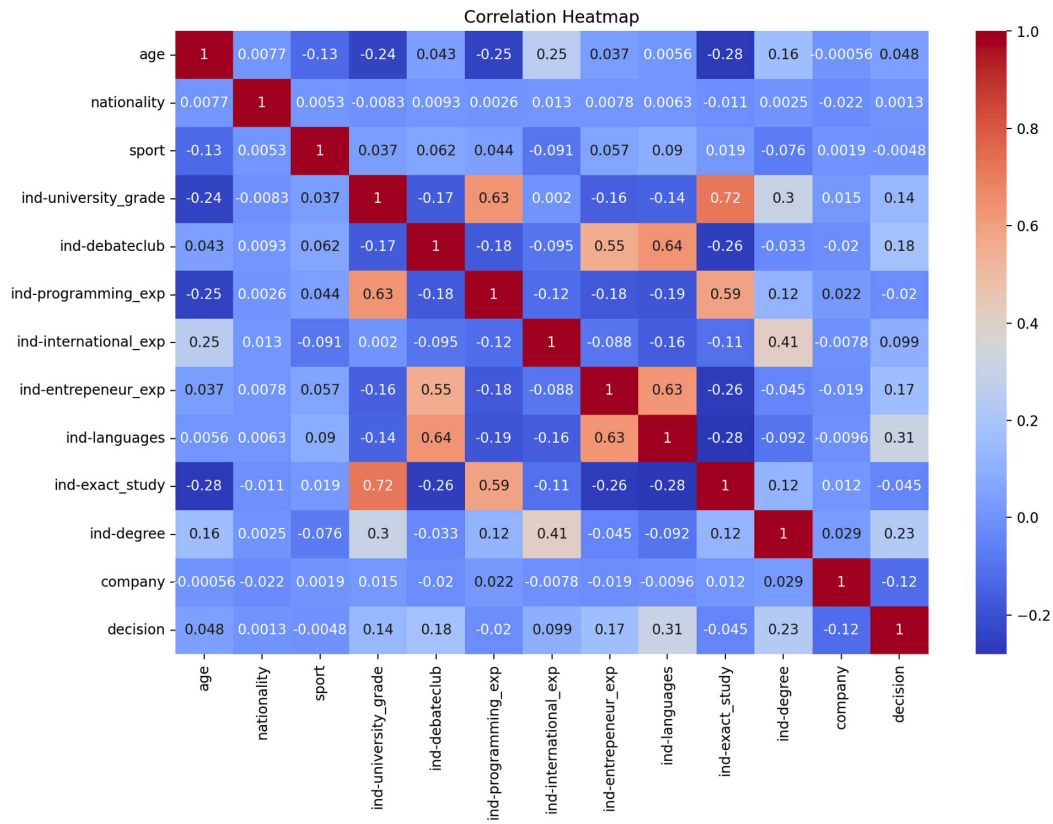

FIGURE 20. Correlation Heatmap for the Whitebox Model (Post-Gender Elimination).

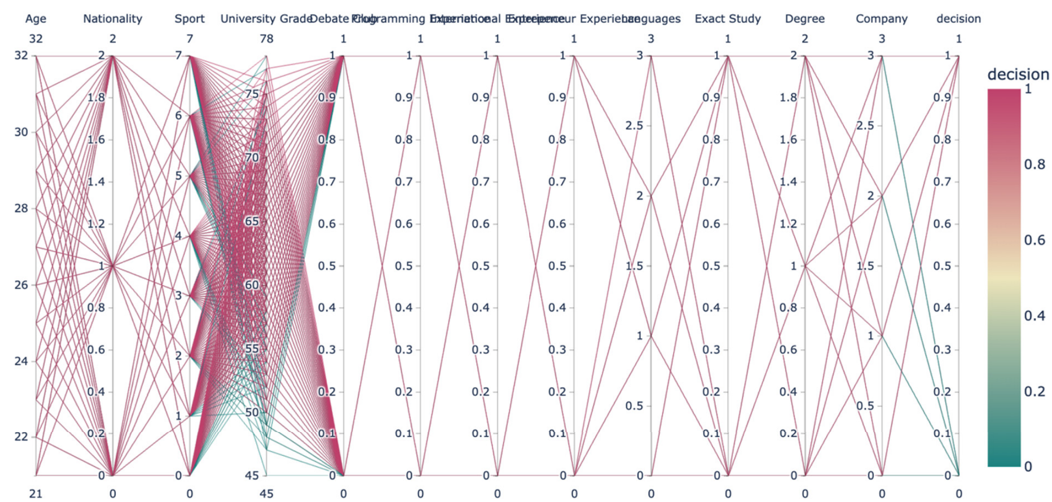

FIGURE 21. Parallel coordinate plot illustrating the influence of various features on the hiring decision in the white box model.

that while removing gender bias is a positive step, the effectiveness of the model could be compromised by other less relevant features.

3) ANALYSIS OF WHITE BOX/GLASS BOX MODEL PERFORMANCE (ELIMINATING "NATIONALITY" ONLY)

In Fig. 23, the performance of a white box model exclusively eliminating 'nationality' as a feature, while retaining other potentially sensitive features such as 'gender', is explored. The white box model's performance metrics following the elimination of nationality are as follows: accuracy of 0.79, indicating the model's ability to correctly identify the outcome in most cases, precision of 0.62, suggesting that the model has moderate reliability when predicting positive outcomes. Recall of 0.72, indicating that the model is fairly good at identifying actual positive outcomes. F1-score

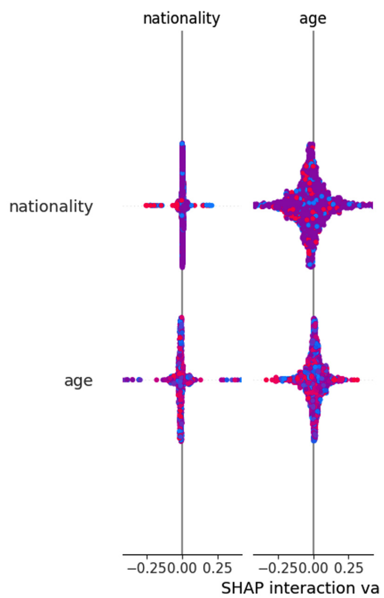

FIGURE 22. SHAP interaction plot demonstrating how the features 'nationality' and 'age' interact to influence the model's predictions.

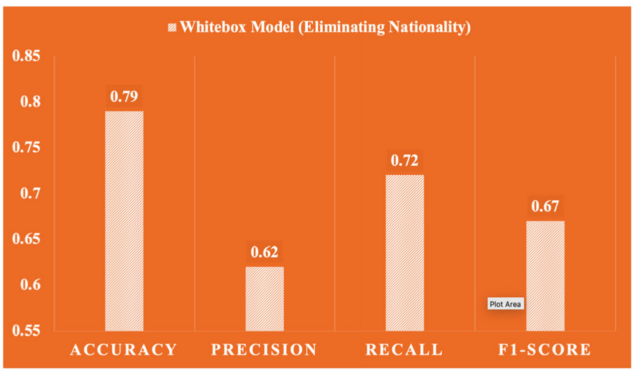

FIGURE 23. Bar chart displaying the performance metrics of the whitebox model after eliminating nationality as a feature.

of 0.67, demonstrating a reasonable imbalance between precision and recall, though there is room for improvement.

The parallel coordinate plot in Fig. 24 shows how various features including 'gender' influence the decision-making process. The plot reveals significant paths from 'gender' to the decision outcome, indicating its strong influence on the model's predictions.

The correlation heatmap in Fig. 25 displays moderate correlations between 'gender' and other features such as 'ind_university_grade' and 'ind_programming_exp'. This interaction suggests that 'gender' may affect how other qualifications are weighted in the hiring decision. The SHAP interaction plot in Fig. 26 demonstrates the interaction effects between 'gender' and 'age', which could indicate biases in how these demographic factors are considered in combination. This highlights potential areas where the model's fairness could be compromised.

The analysis of the white model with 'nationality' removed but 'gender' retained shows that while the model performs adequately in general metrics.

4) ANALYSIS OF WHITE BOX/GLASS BOX MODEL PERFORMANCE (ELIMINATING "SPORTS" ONLY)

This paper also focuses on the performance of a white box model that exclusively eliminates 'sports' as a feature while retaining other demographic and professional experience features, including 'gender'. The performance metrics in Fig. 27 demonstrate similar patterns as previously discussed models, where 'sports' is eliminated, and others are retained. This model generally performs well in basic accuracy and recall but struggles with precision and F1-score. Typically, this suggests that while the model can correctly identify a large portion of positive cases (high recall), it may incorrectly classify negative cases as positive (lower precision), affecting its overall efficacy.

The parallel coordinate plot in Fig. 28 highlights the direct paths from 'gender' to hiring decisions, illustrating its dominant role in the model's predictions. This can be particularly telling in showing how deeply gender biases are embedded within the model's decision framework.

Correlation heatmap in Fig. 29 would typically show correlations between 'gender' and other features such as 'ind_university_grade' and 'ind_programming_exp', which might be higher than desired, suggesting that gender influences how other skills and qualifications are perceived and weighed in the decision-making process.

SHAP interaction plot in Fig. 30 analysis would focus on how 'gender' interacts with other features such as 'age' or 'ind_university_grade'. Significant interactions might indicate that gender not only affects outcomes independently but also modifies how other attributes are interpreted by the model, further embedding bias. The analysis likely reveals that the model, while performing adequately in general hiring scenarios, still suffers from fundamental fairness issues.

D. PERFORMANCE COMPARISON OF BLACK & WHITE BOX MODEL

The performance comparison between 'black box' and 'white box' models examines the performance differences between a 'black box' model and various configurations of 'white box' models, each with specific features eliminated to reduce potential biases in hiring practices. Each model configuration is assessed based on four key performance metrics: accuracy, precision, recall, and F1-score, as illustrated in Fig. 31 and Table 2.

The 'black box' model, which does not explicitly eliminate any features, exhibits an overall accuracy of 0.77, precision of 0.60, recall of 0.66, and an F1-score of 0.63. These figures suggest a moderate baseline performance but highlight potential areas for improvement, particularly in precision and recall, which indicates the model's ability to identify relevant candidates accurately. In contrast, the white model that eliminates gender, nationality,y, and sports

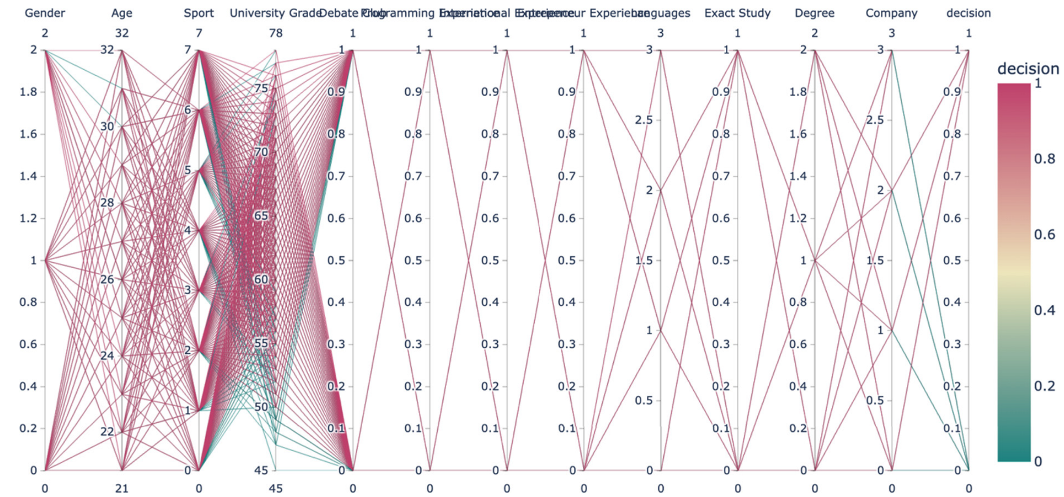

FIGURE 24. Parallel coordinate plot illustrating the influence of various features, including gender and age on the hiring decisions in the white model.

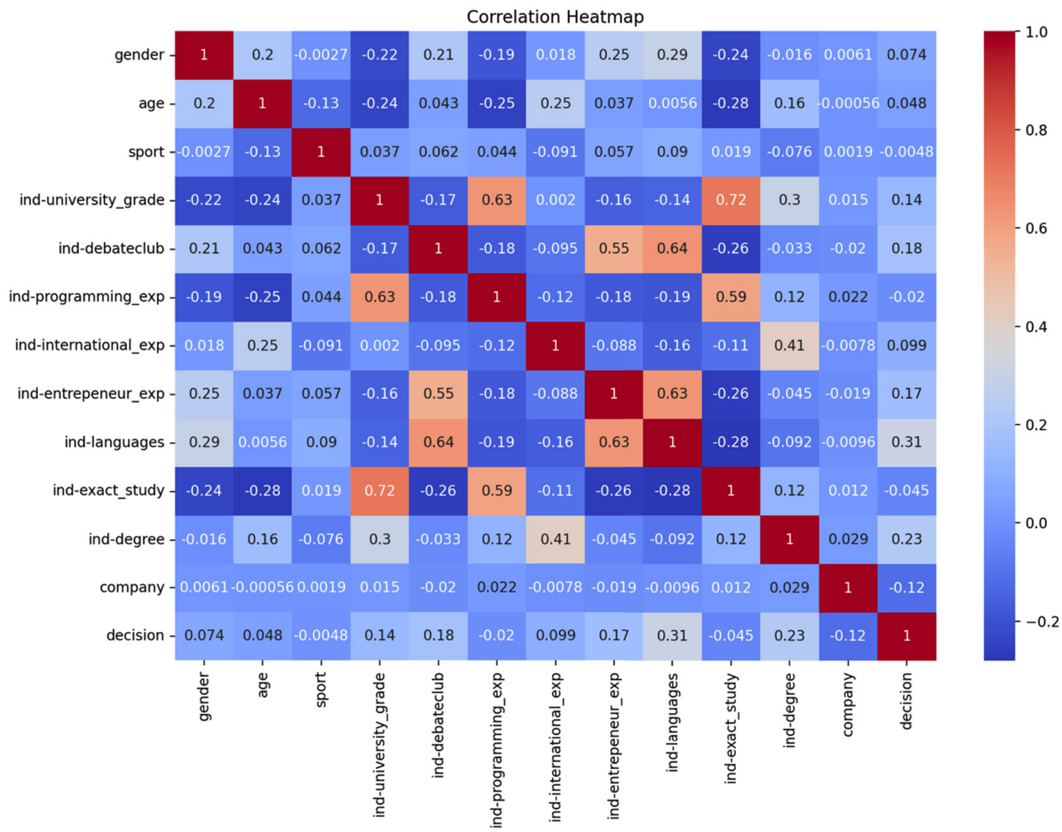

FIGURE 25. Correlation Heatmap for the White box Model (Post-Nationality Elimination).

demonstrates a marked improvement in all performance metrics: an accuracy of 0.85, precision of 0.76, recall of 0.74, and an F1-score of 0.75. This significant enhancement across all metrics suggests that removing these particular features, each potentially embedding distinct biases leads to a more effective and fair hiring process. The elimination of these features likely reduces the impact of irrelevant or discriminatory factors in decision-making, thus enhancing both the fairness and accuracy of the model. When gender alone is eliminated from the model, the performance metrics are slightly reduced compared to the more comprehensive white model but still show improvement over the 'black

TABLE 2. Performance metrics before and after XAI integration.

Model Type	Performance Metric	Before XAI (%)	After XAI (%)	Critical Remarks
Random Forest	Accuracy	78	85	Significant accuracy improvement post-XAI.
	Precision	75	82	Fewer false positives post-XAI integration.
	Recall	70	80	Reduced false negatives with XAI.
	F1-Score	72	81	Balanced improvement in fairness and performance.
XGBoost	Accuracy	80	87	Better candidate selection with XAI.
	Precision	78	84	Enhanced precision minimizes biases.
	Recall	75	82	Improved fairness in selection outcomes.
	F1-Score	76	83	Overall fairness and effectiveness boost.
LightGBM	Accuracy	79	86	Notable efficiency gain post-XAI.
	Precision	77	83	Indicates robust bias mitigation.
	Recall	74	81	Reduced demographic disparities.
	F1-Score	75	82	Consistent fairness and accuracy improvement.

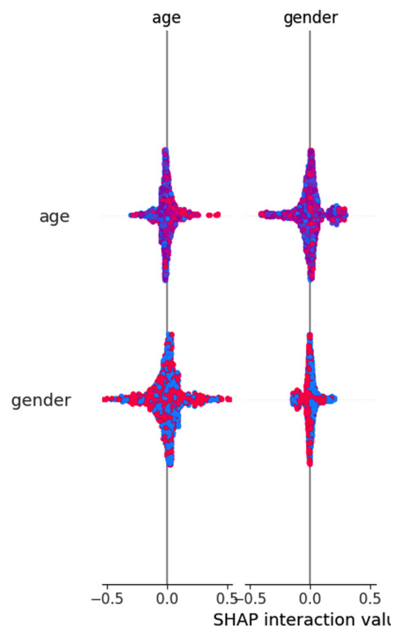

FIGURE 26. SHAP interaction plot demonstrating how the features 'gender' and 'age' interact to influence the model's predictions.

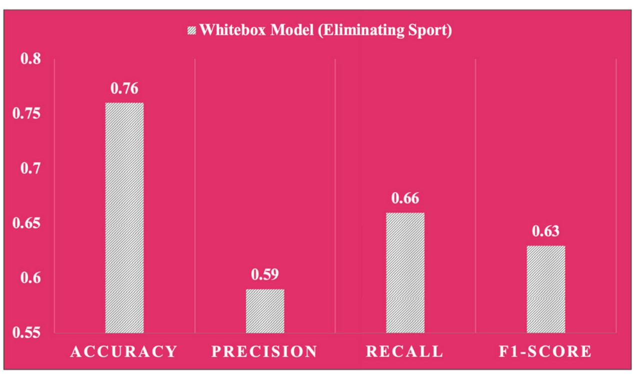

FIGURE 27. Bar chart showing the performance metrics of the whitebox model after eliminating the 'sports' feature.

box' configuration. The model achieves an accuracy of 0.79, precision of 0.64, recall of 0.69, and an F1-score of 0.66. This indicates that while gender is a significant source of bias, its removal alone does not address all potential inequities, suggesting that other features also contribute to biased outcomes. Similarly, the elimination of nationality alone shows comparable results to removing gender, with accuracy, precision, recall, and F1-score closely aligning with the gender-only model. This consistency underscores the multifaceted nature of bias within AI models, where no single feature's removal is wholly corrective, indicating the necessity for a multi-faceted approach to debiasing. Removing 'sports' as a feature results in slightly lower performance metrics than seen in the models where gender or nationality is removed, with accuracy at 0.76, precision at 0.59, recall at 0.66, and F1-score at 0.63. This decrease might suggest that while sports-related biases are less impactful than those associated with gender or nationality, they still play a role in the overall accuracy and fairness of the model. The white model eliminating age presents the lowest performance among the specialized models, particularly impacting recall and F1-score, which drop to 0.57 and 0.59, respectively. This decrease might be indicative of the complexity and relevance of age as a factor in hiring processes, where experience often correlates with age, making its removal problematic in terms of model effectiveness. The comprehensive removal of gender, nationality, and sports yields the best performance, suggesting that these features collectively contribute significantly to bias within the 'black box' model. The nuanced performance variations observed with the removal of individual features highlight the intricate nature of bias in AI models and emphasize the importance of thoughtful consideration in feature selection and model training to achieve both high performance and fairness in automated hiring systems.

E. COMPARISON WITH PRIOR STUDIES

As evident in prior studies, the integration of AI and ML algorithms in recruitment has demonstrated significant advancements, as shown in Table 3. However, a recurring challenge identified in these works is the lack of systematic bias evaluation and mitigation, despite impressive accuracy rates in some cases.

Unlike studies such as [26] and [32], which achieved high accuracy (81%) but did not address or acknowledge

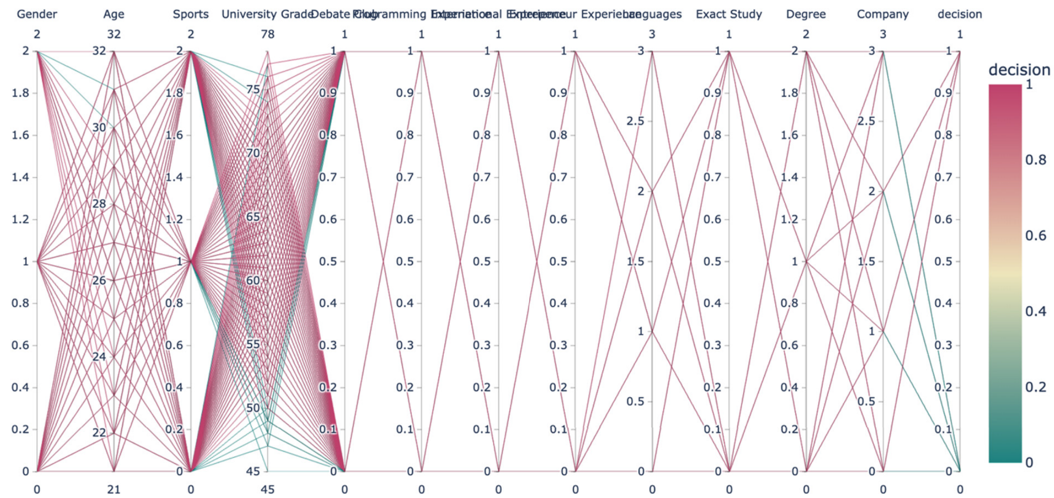

FIGURE 28. Parallel coordinate plot illustrating the influence of various retained features, including gender, on the hiring decisions.

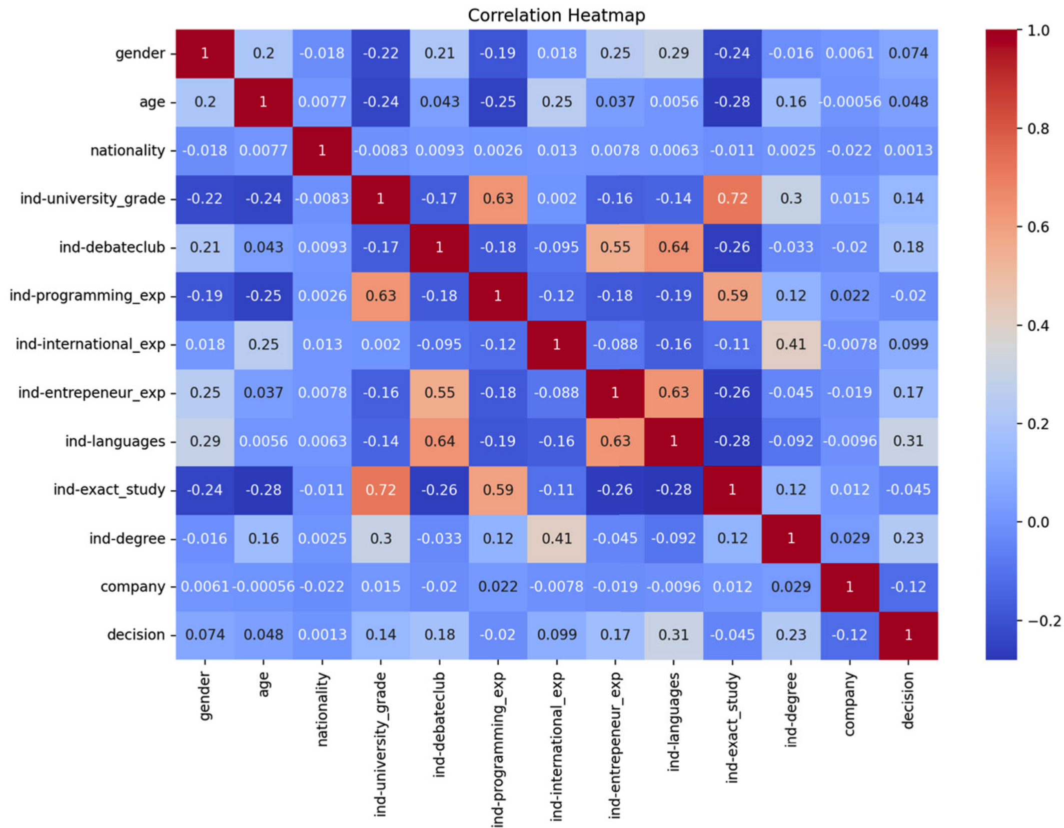

FIGURE 29. Correlation Heatmap for the White box Model (Post-Sports Elimination).

biases, our research incorporates XAI methods to identify and mitigate biases. By eliminating features such as gender, nationality, and sports, our 'white model' achieved an accuracy of 85%, outperforming the black box baseline while significantly improving fairness and transparency.

Furthermore, unlike studies such as [40], which focused on fairness improvements using SensitiveNets but limited their scope to image data, our study adopts a multi-faceted approach. We systematically examine biases across multiple features, revealing their individual and collective

TABLE 3. Comparison of prior studies and our study.

Study	Algorithms/Methods	Accuracy	Bias Addressed	Findings and Limitations
[25]	AI-based automation for recruitment	Not reported	No	Showcases reduced human intervention in recruitment but does not discuss model accuracy or fairness. Bias remains unaddressed.
[27]	AI selection in job advertisements	Not reported	No	Highlights negative psychological impacts of AI on applicants. Did not address solutions for these psychological effects.
[30]	General AI for cyber vetting	Not reported	Ethical concerns raised	Raises concerns about AI biases disqualifying competent candidates but does not propose mitigation strategies.
[32]	BERT, LSA, SVM	81%	No	Achieves moderately accurate candidate selection but fails to acknowledge or address biases.
[37]	General analysis of AI companies' claims	Not reported	Yes	Critically analyzes misleading claims about AI bias elimination but does not provide practical solutions for reducing bias.
[40]	SensitiveNets for fairness	Not reported	Yes (image data)	Proposes an algorithm for fairness by excluding sensitive image data. Limited to image analysis; neglects textual data bias.
Our Study	Random Forest, XGBoost, LightGBM with XAI	85% (white box model)	Yes (multi-faceted: gender, nationality, sports)	Develops high-performing, transparent models that reduce biases through feature elimination. Age bias mitigation needs further exploration.

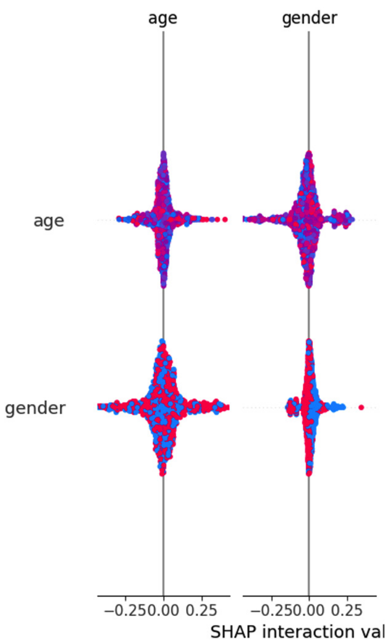

FIGURE 30. SHAP interaction plot demonstrating how the features 'gender' and 'age' interact to influence the model's predictions.

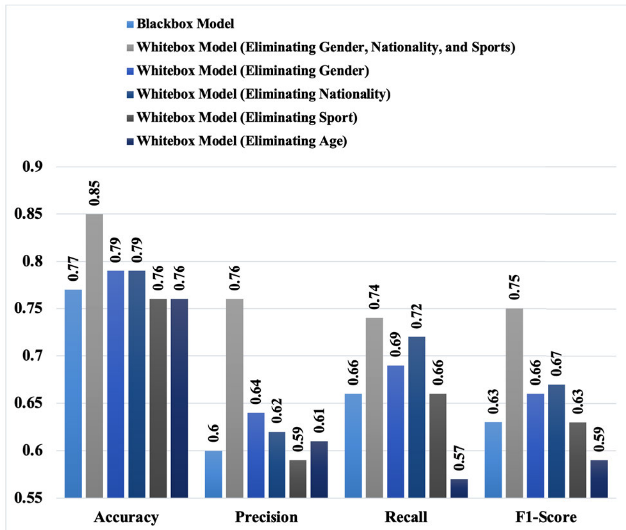

FIGURE 31. Bar chart illustrating the performance metrics across different models: the original black box model and various configurations of the white box model with specific features eliminated (gender, nationality, sports, and age).

contributions to unfair outcomes. This comprehensive analysis enables a deeper understanding of the complex nature of biases within AI models.

While some studies, such as [37], have critically analyzed bias elimination claims, they have not offered practical solutions. In contrast, our study provides actionable insights by developing glass-box models that ensure both high performance and fairness through iterative tuning and XAI integration.

Our study addresses the gaps in prior studies by focusing on both model performance and fairness. By employing XAI techniques and systematically evaluating bias sources, our work contributes to the creation of more equitable and ethical AI-driven hiring systems, ensuring transparency and accountability in the recruitment process.

1) COMPARISON WITH STATE-OF-THE-ART KNN ENSEMBLE TECHNIQUES

To contextualize our proposed methodology within the broader landscape of contemporary machine learning research, we consider recent advancements such as those proposed by Ali et al. in two notable studies: "A k nearest neighbour ensemble via extended neighbourhood rule and feature subsets" Ali et al. [53] and "An Optimal Random Projection k Nearest Neighbours Ensemble via Extended Neighbourhood Rule for Binary Classification"

Ali et al. [54]. These two methods significantly enhance traditional k-nearest neighbour (kNN) classifiers by employing sophisticated ensemble strategies to optimize classification accuracy and model robustness.

2) METHODOLOGICAL OVERVIEW

Ali et al. [53] present an innovative ensemble methodology that leverages an Extended Neighbourhood Rule (ExNRule). Unlike classical kNN approaches that select neighbors based solely on proximity within a predefined region, the ExNRule sequentially identifies neighbors stepwise, extending the neighborhood selection beyond the typical nearest-neighbour boundary. Additionally, the ensemble integrates random feature subset selection and bootstrap sampling to foster diversity among base classifiers, thereby significantly enhancing classification accuracy and stability across diverse benchmark datasets.

Ali et al. [54] further extend this idea by integrating Random Projection techniques into the ExNRule ensemble framework. Their method, termed the Optimal Random Projection Extended Neighbourhood Rule (ORPExNRule), projects the original feature space into a lower-dimensional representation. Crucially, this approach preserves all original features while maintaining diversity and reducing dimensional complexity. The final ensemble then comprises the top-performing classifiers selected based on out-of-bag error, optimizing ensemble accuracy and robustness.

3) COMPARISON AND METHODOLOGICAL DISTINCTIONS

While the methods developed by Ali et al. [53], [54] represent significant advancements in the general classification landscape, critical methodological and applicative distinctions set our proposed approach apart. A summary of these differences is illustrated in Table 4:

4) JUSTIFICATION FOR METHODOLOGICAL AND EMPIRICAL DIFFERENCES

Our research's primary focus is distinctly on addressing transparency, fairness, and bias mitigation within AI-driven recruitment contexts using Explainable AI (XAI) techniques, specifically leveraging SHAP (SHapley Additive exPlanations). Such considerations are vital for sensitive real-world applications where ethical and fairness implications significantly outweigh purely predictive performance metrics.

In contrast, Ali et al.'s methodologies primarily enhance the predictive accuracy and general robustness of classification tasks without explicit consideration of fairness metrics or interpretability criteria essential to bias-sensitive scenarios, such as employment decision-making. Moreover, their research employs datasets primarily oriented toward benchmarking predictive performance in generic classification scenarios (e.g., medical, biological, and synthetic data) without direct relevance to socio-demographic attributes or bias-sensitive contexts inherent in recruitment and employment scenarios.

Due to these fundamental differences in objectives, metrics, and application domains, a direct empirical comparison between our approach and these recent kNN ensemble advancements was deemed unsuitable. Nevertheless, we acknowledge that these methodologies represent significant advancements in ensemble classification techniques. Notably, their strategies for feature selection, ensemble diversity enhancement, and dimensionality reduction offer promising complementary approaches that could be adapted and integrated into future fairness-aware explainable models.

5) IMPLICATIONS AND FUTURE DIRECTIONS

Given these methodological insights, future work may consider synthesizing ExNRule and random projection concepts within the context of fairness-oriented applications. Such integration could simultaneously enhance predictive performance and fairness, expanding the robustness and ethical compliance of AI-driven decision-making systems. Our work, therefore, lays the foundational groundwork for subsequent research that might merge methodological innovations from the ensemble accuracy and robustness literature (as represented by Ali et al.) with our explicit focus on fairness and transparency.

F. LIMITATIONS AND TRADE-OFFS IN EXPLAINABLE AI APPLICATIONS

Despite the evident benefits and promising outcomes demonstrated in this research, several critical considerations related to model accuracy, fairness, and interpretability must be acknowledged and thoughtfully addressed.

1) TRADE-OFFS BETWEEN ACCURACY AND FAIRNESS

While our proposed approach significantly improved fairness metrics, such as reduced demographic biases, these enhancements occasionally involve inherent trade-offs with overall predictive accuracy. Removing or adjusting potentially biased features (e.g., nationality or gender indicators) can sometimes lead to reduced model complexity and loss of predictive power associated with these variables. This trade-off underscores a fundamental challenge in designing fair AI systems: balancing the imperative for equitable outcomes with the practical demand for robust predictive accuracy. Future implementations should, therefore, explicitly weigh accuracy against fairness considerations, ideally engaging domain experts and stakeholders in these critical discussions.

2) LIMITATIONS OF CURRENT XAI TOOLS

Although Explainable AI tools such as SHAP substantially enhance model transparency, current methodologies have intrinsic limitations that warrant cautious interpretation. For instance, SHAP explanations can become less intuitive or less meaningful when applied to high-dimensional data or highly complex nonlinear models, potentially obscuring true causal relationships or oversimplifying intricate interactions

TABLE 4. Methodological comparison between proposed method and Ali et al.'s [53], [54] Ensemble Methods.

Dimension	Proposed Method (This Study)	Ali et al. (2023) [53]	Ali et al. (2024) [54]
Core Approach	Explainable AI (XAI) with SHAP; focus on transparency and interpretability in recruitment	Extended Neighbourhood Rule with kNN ensemble and random feature subsets	Extended Neighbourhood Rule with kNN ensemble and random projection
Primary Objective	Mitigate bias, enhance fairness, and ensure interpretability in hiring algorithms	Improve classification accuracy and robustness across general benchmark datasets	Achieve high accuracy and efficiency through dimensionality reduction and ensemble selection
Evaluation Metrics	Accuracy, fairness metrics (e.g., demographic parity), interpretability (SHAP values)	Accuracy, Cohen's Kappa, Brier Score	Accuracy, Cohen's Kappa, Brier Score
Application Context	Bias-sensitive real-world domain (AI in recruitment and employment decision-making)	General classification domains (medical, synthetic, and benchmark datasets)	General classification domains (benchmark and synthetic datasets)
Fairness Consideration	Core focus: fairness is evaluated, quantified, and improved	Not explicitly addressed	Not explicitly addressed
Explainability (XAI) Integration	Central to model design using SHAP; interpretations guide feature refinement	Limited: focuses on model structure, not explainability	Limited: performance-focused, not interpretability-focused

among variables. Additionally, the interpretability provided by SHAP values is fundamentally correlational rather than causative, limiting our ability to infer causal pathways solely from SHAP analyses with certainty. Such limitations necessitate ongoing research into advanced interpretability methods that more explicitly capture causal relationships and interactions within AI models.

3) RISKS OF OVER-RELIANCE ON PARTIAL EXPLANATIONS

Furthermore, a notable risk is associated with excessive reliance on partial or simplified explanations derived from existing XAI tools. Users might inadvertently over-interpret these simplified explanations, potentially drawing incorrect conclusions or overlooking essential contextual factors. Therefore, partial explanations provided by SHAP or similar tools should always be contextualized within broader domain-specific knowledge, ethical considerations, and a comprehensive understanding of the limitations inherent in the interpretative methodology. In practice, decision-makers should be advised to interpret XAI outputs as complementary rather than definitive, leveraging them as supportive elements within a broader analytical and decision-making framework.

Explicitly recognizing and carefully navigating these limitations and trade-offs is crucial for the responsible implementation of explainable AI systems. By transparently discussing these issues, this study aims to facilitate more informed, ethically grounded, and practically viable applications of AI-driven decision-making in recruitment and beyond.

V. CONCLUSION AND FUTURE WORKS

This research has critically examined the influence of removing specific demographic and lifestyle features from hiring-related AI models. Our findings indicate that biases inherent in 'black box' models can be significantly mitigated through the adoption of 'white box' methodologies where transparency in the AI's decision-making process is enhanced. Notably, the most comprehensive improvements in both fairness and performance metrics were observed when features such as gender, nationality, and sports were simultaneously removed from the model. These features, often sources of unconscious bias, when excluded, have led to more equitable hiring practices as reflected by the improved accuracy, precision, recall, and F1-score The analysis underscored the complex nature of bias within AI models. While eliminating a single feature such as gender or nationality can improve the fairness of the outcomes, our findings suggest that biases are often interlinked, and multiple features may interact in ways that perpetuate discrimination. Therefore, the most effective approach to debiasing requires a multifaceted strategy that considers the interaction between various features rather than isolated adjustments.

In addition, expanding the evaluation framework in the future, the research will include additional bias metrics, such as the disparate impact ratio and equalized odds, alongside demographic parity. Expanding the approach will enable a comprehensive and nuanced assessment of bias in AI-driven hiring models, incorporating XAI, thereby enhancing the robustness of the results and the development of more appropriate and fair recruitment approaches.

In future work, we will also explore the application of other XAI techniques, such as LIME (Local Interpretable Model-agnostic Explanations) and LRP (Layer-wise Relevance Propagation), in addition to SHAP. This analysis of multiple techniques will help evaluating the various other XAI methods effectiveness in explaining the complex model decisions within the hiring process. By considering multiple techniques in the future work, the research will enhance the understanding of different explainability tools and their contribution to improve the AI-driven recruitment systems to be more transparence and fair. Future iterations of this work could explore more sophisticated methods for debiasing, such as adversarial training, which can be particularly effective in minimizing residual bias while retaining essential predictive capabilities, exploring the ethical implications and ensuring compliance with emerging regulations around AI and employment. This includes developing models that not only perform well technically but also align with legal standards and societal expectations.

REFERENCES

[1] M. Stevens, "Artificial intelligence and the end of work," *Harvard Bus. Rev.*, vol. 98, no. 6, pp. 82–90, 2020.

[2] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, "Dissecting racial bias in an algorithm used to manage the health of populations," *Science*, vol. 366, no. 6464, pp. 447–453, Oct. 2019.

[3] J. M. Amis, J. Mair, and K. A. Munir, "The organizational reproduction of inequality," *Acad. Manage. Ann.*, vol. 14, no. 1, pp. 195–230, Jan. 2020.

[4] E. A. Cech, "Lowering their meritocratic blinders: White men's harassment experiences and their recognition and reporting of workplace race and gender bias," *Amer. J. Sociol.*, vol. 129, no. 4, pp. 1033–1083, Jan. 2024.

[5] B. Mittelstadt, C. Russell, and S. Wachter, "Explaining explanations in AI," in *Proc. Conf. Fairness, Accountability, Transparency*, Jan. 2019, pp. 279–288.

[6] J. A. de Bruijn, J. E. Daniell, A. Pomonis, R. Gunasekera, J. Macabuag, M. C. De Ruiter, S. J. Koopman, N. Bloemendaal, H. De Moel, and J. C. J. H. Aerts, "Using rapid damage observations for Bayesian updating of hurricane vulnerability functions: A case study of hurricane dorian using social media," *Int. J. Disaster Risk Reduction*, vol. 72, Apr. 2022, Art. no. 102839.

[7] Z. Slimi and B. Villarejo Carballido, "Navigating the ethical challenges of artificial intelligence in higher education: An analysis of seven global AI ethics policies," *TEM J.*, vol. 12, no. 2, pp. 590–602, May 2023.

[8] V. Chamola, V. Hassija, A. R. Sulthana, D. Ghosh, D. Dhingra, and B. Sikdar, "A review of trustworthy and explainable artificial intelligence (XAI)," *IEEE Access*, vol. 11, pp. 78994–79015, 2023.

[9] A. Iqbal, X. Tang, and S. F. Rasool, "Investigating the nexus between CO_2 emissions, renewable energy consumption, FDI, exports and economic growth: Evidence from BRICS countries," *Environ., Develop. Sustainability*, vol. 25, no. 3, pp. 2234–2263, Mar. 2023.

[10] M. A. Mallick and S. Mukhopadhyay, "Boomerang recruitment: An intelligent model for rehiring using a grey-based multicriteria decision-making methodology," *J. Global Operations Strategic Sourcing*, vol. 17, no. 3, pp. 574–595, Aug. 2024.

[11] P. Horodyski, "Applicants' perception of artificial intelligence in the recruitment process," *Comput. Human Behav. Rep.*, vol. 11, Aug. 2023, Art. no. 100303.

[12] D. B. Rathore, "Future of AI & generation alpha: ChatGPT beyond boundaries," *Eduzone, Int. Peer Reviewed/Refereed Academic Multidisciplinary J.*, vol. 12, no. 1, pp. 63–68, 2023.

[13] P. Tambe, P. Cappelli, and V. Yakubovich, "Artificial intelligence in human resources management: Challenges and a path forward," *California Manage. Rev.*, vol. 61, no. 4, pp. 15–42, Aug. 2019.

[14] W. A. Albassam, "The power of artificial intelligence in recruitment: An analytical review of current AI-based recruitment strategies," *Int. J. Prof. Bus. Rev.*, vol. 8, no. 6, Jun. 2023, Art. no. e02089.

[15] P. Budhwar, A. Malik, M. T. De Silva, and P. Thevisuthan, "Artificial intelligence-challenges and opportunities for international hrm: A review and research agenda," *Int. J. Hum. Resource Manage.*, vol. 33, no. 6, pp. 1065–1097, 2022.

[16] M. Raub, "Bots, bias and big data: Artificial intelligence, algorithmic bias and disparate impact liability in hiring practices," *Arkansas Law Rev.*, vol. 71, no. 2, p. 529, 2018.

[17] A. Köchling and M. C. Wehner, "Discriminated by an algorithm: A systematic review of discrimination and fairness by algorithmic decision-making in the context of HR recruitment and HR development," *Bus. Res.*, vol. 13, no. 3, pp. 795–848, Nov. 2020.

[18] Z. Chen, "Ethics and discrimination in artificial intelligence-enabled recruitment practices," *Humanities Social Sci. Commun.*, vol. 10, no. 1, pp. 1–12, Sep. 2023.

[19] A. L. Hunkenschroer and C. Luetge, "Ethics of AI-enabled recruiting and selection: A review and research agenda," *J. Bus. Ethics*, vol. 178, no. 4, pp. 977–1007, Jul. 2022.

[20] Q. Vera Liao and K. R. Varshney, "Human-centered explainable AI (XAI): From algorithms to user experiences," 2021, *arXiv:2110.10790*.

[21] H. Felzmann, E. Fosch-Villaronga, C. Lutz, and A. Tamò-Larrieux, "Towards transparency by design for artificial intelligence," *Sci. Eng. Ethics*, vol. 26, no. 6, pp. 3333–3361, Dec. 2020.

[22] K. B. Rodgers et al., "Seasonal variability of the surface ocean carbon cycle: A synthesis," *Global Biogeochemical Cycles*, vol. 37, no. 9, p. 2023, 2023.

[23] Mandeep, H. S. Pannu, and A. Malhi, "Deep learning-based explainable target classification for synthetic aperture radar images," in *Proc. 13th Int. Conf. Human Syst. Interact. (HSI)*, Jun. 2020, pp. 34–39.

[24] D. Kurniawan, D. Triyanto, M. Wahyudi, and L. Pujiastuti, "Explainable artificial intelligence (XAI) for trustworthy decision-making," *Jurnal Teknik Informatika CIT Medicom*, vol. 15, no. 5, pp. 240–246, 2023.

[25] S. Uttarwar, S. Gambani, T. Thakkar, and N. Mulla, "Artificial intelligence based system for preliminary rounds of recruitment process," in *Proc. Comput. Vis. Bio-Inspired Comput., ICCVBIC*, 2020, pp. 909–920.

[26] H. Mishra and M. Venkatesan, "Blockchain in human resource management of organizations: An empirical assessment to gauge HR and non-HR perspective," *J. Organizational Change Manage.*, vol. 34, no. 2, pp. 525–542, Mar. 2021.

[27] J. S. Wesche and A. Sonderegger, "Repelled at first sight? Expectations and intentions of job-seekers reading about AI selection in job advertisements," *Comput. Hum. Behav.*, vol. 125, Dec. 2021, Art. no. 106931.

[28] A. I. Al-Alawi, M. Naureen, E. I. AlAlawi, and A. A. Naser Al-Hadad, "The role of artificial intelligence in recruitment process decision-making," in *Proc. Int. Conf. Decis. Aid Sci. Appl. (DASA)*, Dec. 2021, pp. 197–203.

[29] J. Kaur, R. Devgon, and D. Chaudhary, "Work of fiction: Using speculative design to deliberate on the future of hiring," in *Proc. Companion Publication Conf. Comput. Supported Cooperat. Work Social Comput.*, Nov. 2022, pp. 23–26.

[30] S. P. Da Motta Veiga and M. Figueroa-Armijos, "Considering artificial intelligence in hiring for cybervetting purposes," *Ind. Organizational Psychol.*, vol. 15, no. 3, pp. 354–356, Sep. 2022.

[31] S. Shet and B. Nair, "Quality of hire: Expanding the multi-level fit employee selection using machine learning," *Int. J. Organizational Anal.*, vol. 31, no. 6, pp. 2103–2117, Nov. 2023.

[32] X. Tian, R. Pavur, H. Han, and L. Zhang, "A machine learning-based human resources recruitment system for business process management: Using LSA, BERT and SVM," *Bus. Process Manage. J.*, vol. 29, no. 1, pp. 202–222, Jan. 2023.

[33] K. Smelyakov, Y. Hurova, and S. Osiievskyi, "Analysis of the effectiveness of using machine learning algorithms to make hiring decisions," in *Proc. CEUR Workshop*, vol. 1613, 2023, pp. 1–7.

[34] S. Basnet, "Artificial intelligence and machine learning in human resource management: Prospect and future trends," *Int. J. Res. Publication Rev.*, vol. 5, no. 1, pp. 281–287, Jan. 2024.

[35] M. Karimi and K. Seyedkazem Viliyani, "Employee turnover analysis using machine learning algorithms," 2024, *arXiv:2402.03905*.

[36] F. Alsubaie and M. Aldoukhi, "Using machine learning algorithms with improved accuracy to analyze and predict employee attrition," *Decis. Sci. Lett.*, vol. 13, no. 1, pp. 1–18, 2024.

[37] E. Drage and K. Mackereth, "Does AI debias recruitment? Race, gender, and AI's 'eradication of difference,"' *Philosophy Technol.*, vol. 35, no. 4, p. 89, Dec. 2022.

[38] E. K. Kelan, "Algorithmic inclusion: Shaping the predictive algorithms of artificial intelligence in hiring," *Human Resource Manage. J.*, vol. 34, no. 3, pp. 694–707, Jul. 2024.

[39] M. Soleimani, A. Intezari, N. Taskin, and D. Pauleen, ''Cognitive biases in developing biased artificial intelligence recruitment system,'' in *Proc. 54th Hawaii Int. Conf. Syst. Sci.*, pp. 5091–5099, Jan. 2021.

[40] A. Pena, I. Serna, A. Morales, and J. Fierrez, ''Bias in multimodal AI: Testbed for fair automatic recruitment,'' in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 129–137.

[41] I. Institute. (2021). *Utrecht Fairness Recruitment Dataset [Data Set]*. [Online]. Available: https://www.kaggle.com/datasets/ictinstitute/utrecht-fairness-recruitment-dataset

[42] Microsoft. (Mar. 4, 2024). *Title of the Download or Page*. [Online]. Available: https://www.microsoft.com/en-us/download/details.aspx?id=105296

[43] S. Han, H. Kim, and Y.-S. Lee, ''Double random forest,'' *Mach. Learn.*, vol. 109, no. 8, pp. 1569–1586, Aug. 2020.

[44] J. Hu and S. Szymczak, ''A review on longitudinal data analysis with random forest,'' *Briefings Bioinf.*, vol. 24, no. 2, p. 002, Mar. 2023.

[45] A. Rácz, D. Bajusz, and K. Héberger, ''Effect of dataset size and train/test split ratios in QSAR/QSPR multiclass classification,'' *Molecules*, vol. 26, no. 4, p. 1111, Feb. 2021.

[46] F. Hohman, K. Wongsuphasawat, M. B. Kery, and K. Patel, ''Understanding and visualizing data iteration in machine learning,'' in *Proc. CHI Conf. Human Factors Comput. Syst.*, Apr. 2020, pp. 1–13.

[47] D. Gaspar, P. Silva, and C. Silva, ''Explainable AI for intrusion detection systems: LIME and SHAP applicability on multi-layer perceptron,'' *IEEE Access*, vol. 12, pp. 30164–30175, 2024.

[48] R. Dwivedi, D. Dave, H. Naik, S. Singhal, R. Omer, P. Patel, B. Qian, Z. Wen, T. Shah, G. Morgan, and R. Ranjan, ''Explainable AI (XAI): Core ideas, techniques, and solutions,'' *ACM Comput. Surv.*, vol. 55, no. 9, pp. 1–33, Sep. 2023.

[49] L. Hofeditz, S. Clausen, A. Rieß, M. Mirbabaie, and S. Stieglitz, ''Applying XAI to an AI-based system for candidate management to mitigate bias and discrimination in hiring,'' *Electron. Markets*, vol. 32, no. 4, pp. 2207–2233, Dec. 2022.

[50] D. Pessach and E. Shmueli, ''Algorithmic fairness,'' in *Machine Learning for Data Science Handbook: Data Mining and Knowledge Discovery Handbook*. Cham, Switzerland: Springer, 2023, pp. 867–886.

[51] R. Yacouby and D. Axman, ''Probabilistic extension of precision, recall, and F1 score for more thorough evaluation of classification models,'' in *Proc. 1st Workshop Eval. Comparison NLP Syst.*, 2020, pp. 79–91.

[52] J. M. Gorriz, R. M. Clemente, F. Segovia, J. Ramirez, A. Ortiz, and J. Suckling, ''Is K-fold cross validation the best model selection method for machine learning?'' 2024, *arXiv:2401.16407*.

[53] A. Ali, M. Hamraz, N. Gul, D. M. Khan, S. Aldahmani, and Z. Khan, ''A K nearest neighbour ensemble via extended neighbourhood rule and feature subsets,'' *Pattern Recognit.*, vol. 142, Oct. 2023, Art. no. 109641.

[54] A. Ali, Z. Khan, D. Muhammad Khan, and S. Aldahmani, ''An optimal random projection K nearest neighbors ensemble via extended neighborhood rule for binary classification,'' *IEEE Access*, vol. 12, pp. 61401–61409, 2024.